

Contents

1 Project Overview	3
2 Project Background	6
2.1 Clinical Burden of Radiotherapy-Related Intestinal Injury	6
2.2 Main Characteristics of Radiation-Related Intestinal Injury	7
2.3 Bottlenecks of Existing Therapeutic Approaches	8
2.4 Policy and Strategic Responses	8
3 Pain Points of Existing Protection and Intervention Regimens	9
3.1 Existing Radiation Protection and Medical Intervention Regimens	9
3.2 Limitations of Existing Regimens and Unmet Needs	9
4 Market Size Forecast	11
4.1 Global Market Size	11
4.2 Regional Market Size Analysis	13
5 Market Analysis	15
5.1 PEST Analysis	15
5.1.1 Political Analysis	15
5.1.2 Economic Analysis	16
5.1.3 Social Analysis	17
5.1.4 Technological Analysis	18
5.2 SWOT Analysis	20
5.2.1 Strengths	20
5.2.2 Weaknesses	20
5.2.3 Opportunities	21
5.2.4 Threats	21
6 Commercialization Path	21
6.1 Technology and Intellectual Property Protection	21
6.1.1 Intellectual Property	22
6.1.2 Confidentiality Measures	22
6.1.3 Rapid Market Conversion	22
6.1.4 Disclaimer and Phased Statements	23
6.2 Value Chain Analysis	23
6.3 Commercialization Strategy	24
6.3.1 Commercialization Strategy Environment	24
6.3.2 Partners and Resources	25
6.3.3 Commercialization Advancement Steps	26
7 Development Strategy	27

7.1 STP Analysis—Market Segmentation	27
7.1.1 Asia-Pacific Market	27
7.1.2 North American Market	28
7.1.3 European Market	29
7.2 STP Analysis—Target Market Selection	30
7.3 STP Analysis—Market Positioning	31
7.4 Marketing	32
7.4.1 Marketing Objectives	32
7.4.2 Marketing Strategies	32
8 MVP	34
8.1 MVP Design	34
8.2 MVP Validation	35
8.3 Preliminary Cooperation Intentions	35
9 Financial Planning	35
9.1 Cost Assumptions	35
9.2 Revenue Assumptions	39
9.3 Financing Plan	40
9.3.1 Financing Benchmarks	40
9.3.2 Financing Proposal	40
9.3.3 Corporate Valuation	41
9.3.4 Target Financing Capital	42
9.4 Cash Flow Forecast	42
10 Risk Control	43
10.1 Risk Interview	43
10.2 Risk Identification and Assessment	45
10.3 Risk Management Measures	46
10.4 Special Response Measures for High-Risk Matters	46
10.4.1 Development Window Risk	46
10.4.2 Approval Risk	47
10.4.3 Product-Specific Risk	47
10.4.4 Cash Flow Risk	48
10.4.5 Ethical Risk	48
11 CCiC and Industry Collaboration	49
11.1 CCiC Background and Project Participation Positioning	49
11.2 Value of Project Participation in CCiC	49
11.3 CCiC's Role in Advancing the Commercialization Path	50
11.4 Future Industry Cooperation and Development Prospects	51

1 Project Overview

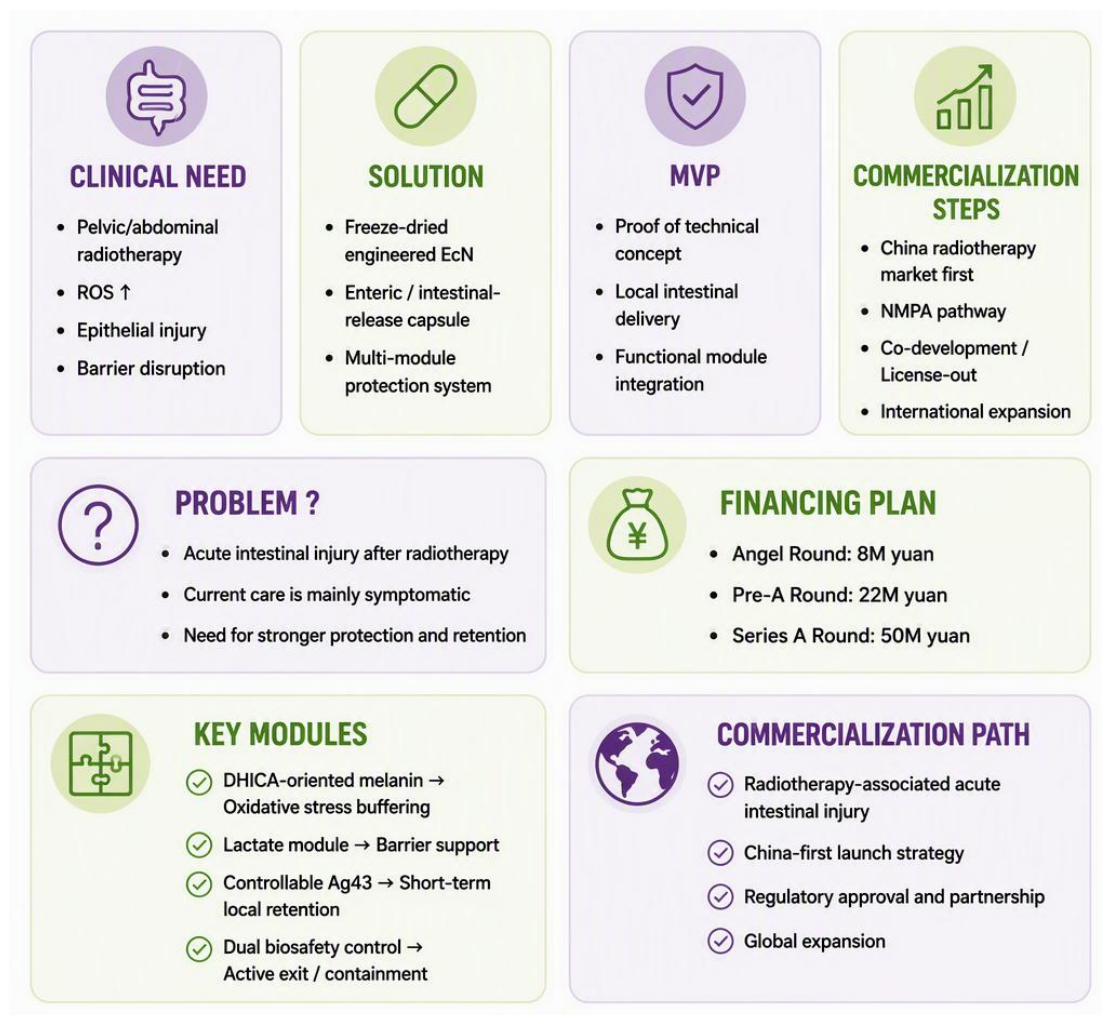


Figure 1 Sybio-Gutweaver Project Overview

Radiotherapy for abdominopelvic tumors can cause oxidative stress, mucosal injury, and intestinal barrier disruption, and clinical management remains predominantly symptomatic and supportive. Sybio-Gutweaver is developing an oral live bacterial candidate formulation based on engineered *Escherichia coli* Nissle 1917 (EcN), targeting radiotherapy-related acute intestinal injury.

The candidate product is intended to use lyophilized live bacteria and enteric-release capsules, integrating a DHICA component-optimized melanin antioxidant module, a lactate barrier support module, an Ag43-controlled adhesion module, and dual biosafety switches. Ag43 is used to enhance short-term retention in the target intestinal segment rather than to pursue long-term colonization; the dual safety mechanisms respectively employ high lactate-triggered Qwitch self-destruction and L-arabinose-dependent hok/sok survival licensing linked to Ag43 expression, reducing the risk of continuous output in abnormal microenvironments and dissemination after departure from the preset scenario.

Business Model

The project adopts an asset-light model of “independent R&D + joint development/technology licensing”, focusing on core strains, functional circuits, safety control, delivery systems, and intellectual property, and realizes returns through technology licensing, milestone payments, and sales royalties after key milestones.

R&D and Regulatory Strategy

The project adopts a regulatory strategy of “NMPA primary pathway, FDA parallel evaluation”:

- China: positioned as a therapeutic live engineered bacterial drug, advancing in accordance with NMPA drug registration requirements, and at key R&D milestones clarifying the specific registration classification and application dossier requirements through CDE communications.
- United States: pursuing parallel planning under the FDA Live Biotherapeutic Product (LBP) framework, focusing on establishing IND/CMC materials and data, including strain identification, manufacturing process, potency, quality control, and stability.

Licensing and Collaboration Strategy

To reduce R&D and industrialization risks, the project adopts a phased collaboration model:

Early-stage R&D phase: Relying on university research platforms to complete the construction of engineered EcN strains, functional validation, and product prototype development, laying the foundation for subsequent preclinical research.

Later-stage translation phase: Collaborating with hospitals, research institutions, and pharmaceutical companies to advance clinical trials, formulation development, scale-up manufacturing, and regulatory submissions, and promoting product

translation through joint development or technology licensing.

At the current stage, the project has completed preliminary demand and product direction validation through interviews with hospitals, research institutions, and industry experts. The interviewees include Gansu Provincial Hospital, the Heavy Ion Center of Wuwei Tumor Hospital of Gansu Province, the Institute of Modern Physics, Chinese Academy of Sciences, and industry experts in live bacterial formulations, among others, providing a demand basis for subsequently establishing clinical research, formulation process, regulatory translation, and industrial licensing collaborations.

Market Strategy

The project adopts the market strategy of “China ground-based radiotherapy first, parallel international market layout, and long-term extension to space scenarios”:

Ground-based radiotherapy as the lead market: long-term deep-space missions have high strategic value, but the near-term target population is limited, and validation and application cycles are relatively long. Therefore, priority is given to establishing a clinical and commercial closed loop based on radiotherapy-related acute intestinal injury in abdominopelvic tumors, and gradually expanding from Northwest China, where there is an existing clinical resource base, to major tumor and radiotherapy centers nationwide.

International and space markets: in parallel, reserve an international registration interface in accordance with FDA LBP requirements. After sufficient product and clinical evidence is generated in ground medical scenarios, further expand into international markets and long-term deep-space radiation health protection applications.

Financing Plan

The financing roadmap corresponds to product R&D milestones and is divided into three stages:

- Angel round: RMB 10 million. Complete the integration of core functional modules, Ag43, and dual biosafety switches; form a lyophilized enteric capsule minimum viable product (MVP); and advance intellectual property layout.
- Pre-A round: RMB 25 million. Complete candidate strain convergence, functional and safety metrics, genetic and formulation stability, critical quality attributes, and manufacturing process development, and advance NMPA and FDA registration preparation.
- Series A: RMB 55 million. Advance registration-grade manufacturing and quality systems, clinical submission, and early clinical development, and initiate co-development and license-out with pharmaceutical companies.

Total planned financing is RMB 90 million, of which about RMB 83 million is for R&D and operations, and RMB 7 million serves as a cash safety reserve.

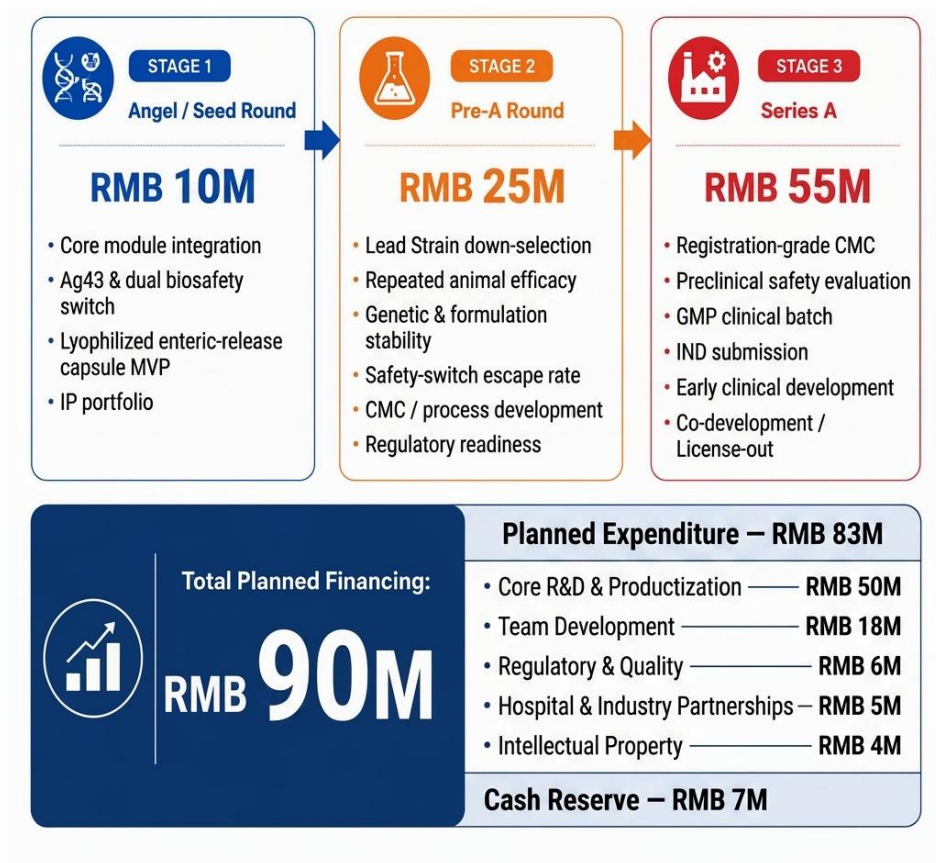


Figure 2 Financing and Development Plan

Strategic Summary

Sybio-Gutweaver integrates antioxidant activity, barrier support, controllable adhesion, active biosafety, and intestinal delivery into a unified engineered live bacterial system. With ground-based radiotherapy as the lead application scenario, it forms a commercial path extending from engineering design to clinical translation and industrial licensing.

2 Project Background

2.1 Clinical Burden of Radiotherapy-Related Intestinal Injury

Large patient base: In 2022, there were approximately 4.8247 million new cases of malignant tumors in China (Han et al., 2024). A national radiotherapy industry survey showed that in 2018, mainland China had 1,463 radiotherapy units, with approximately 1.2596 million patient visits treated annually, creating sustained demand for radiotherapy supportive care (Zhang Ye et al., 2020).

Relatively high acute gastrointestinal toxicity: Approximately 60%-80% of patients receiving abdominal and pelvic radiotherapy may develop acute gastrointestinal toxicity, including diarrhea, abdominal pain, nausea, and abnormal defecation. Intestinal crypts and epithelial cells turn over rapidly and are highly sensitive to

ionizing radiation; therefore, normal intestine is an important dose-limiting tissue in abdominopelvic radiotherapy.

Impact on treatment continuity : More severe acute intestinal injury can lead to dehydration, electrolyte imbalance, nutritional decline, and reduced treatment tolerance. Some patients need to suspend or adjust radiotherapy, which further affects quality of life and treatment completion. Some gastrointestinal functional abnormalities may also persist after treatment completion, making them part of full-cycle cancer management.

2.2 Main Characteristics of Radiation-Related Intestinal Injury

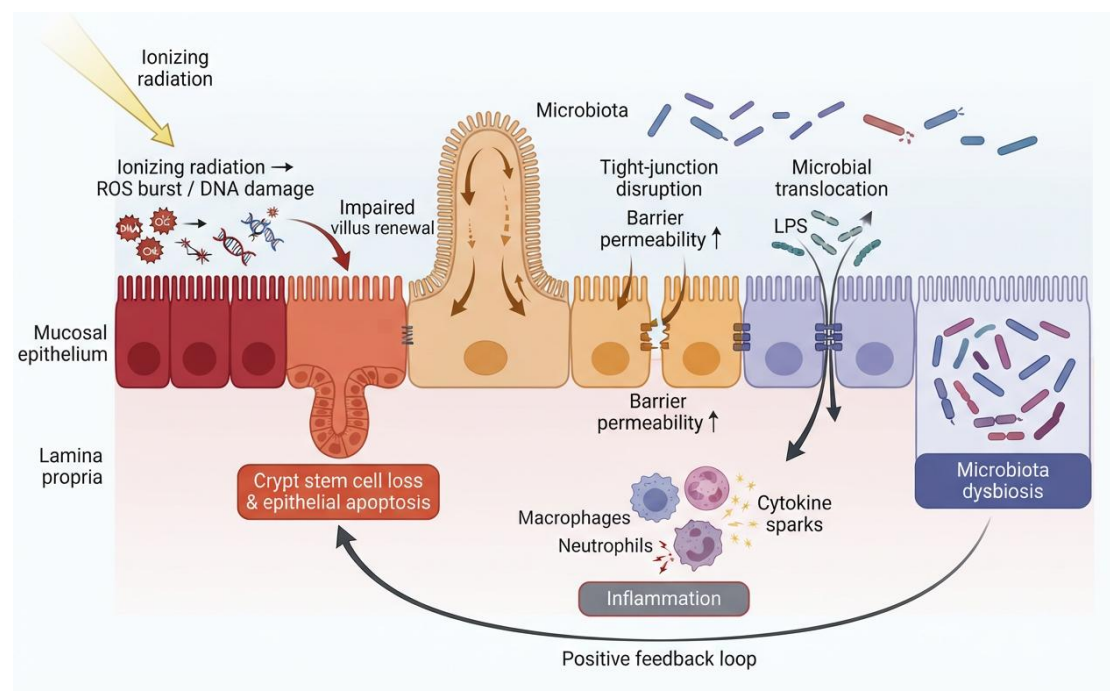


Figure 3 Schematic of the Pathological Process of Radiotherapy-Related Intestinal Injury

Oxidative and mucosal injury: Ionizing radiation can directly damage DNA and also induce large amounts of reactive oxygen species (ROS). After rapidly proliferating intestinal crypt stem cells and epithelial cells are damaged, villus renewal becomes insufficient, and changes such as mucosal inflammation and malabsorption occur (Hauer-Jensen et al., 2014).

Barrier function disruption : Epithelial injury and disruption of intercellular junctions increase intestinal permeability and may promote the translocation of microorganisms and their products across the damaged barrier. Persistent inflammation further hinders barrier recovery, causing symptoms and local tissue injury to amplify each other.

Microecological alterations: Pelvic radiotherapy may be accompanied by changes in gut microbiota diversity and composition, some of which are associated with diarrhea

and radiation enteritis; however, different studies are affected by diet, concurrent chemotherapy, and patients' baseline status, showing marked heterogeneity (Bai et al., 2021).

The above processes are not independent of one another: oxidative stress can exacerbate epithelial injury, and barrier disruption can in turn alter the local microenvironment and microbial homeostasis. Therefore, controlling a single symptom such as diarrhea alone makes it difficult to cover the complete pathological process or provide sustained local mechanistic intervention.

2.3 Bottlenecks of Existing Therapeutic Approaches

1. Radiotherapy techniques cannot completely avoid normal intestinal exposure

Precision radiotherapy such as IMRT and IGRT can reduce doses to normal tissues, but abdominopelvic tumors are anatomically close to the small intestine, colon, and rectum, making exposure of normal intestine difficult to completely avoid. In NRG/RTOG 1203, IMRT reduced gastrointestinal toxicity, but 33.7% of patients still reported frequent or persistent diarrhea at the end of treatment (Klopp et al., 2018).

2. Clinical intervention remains predominantly symptom control

Existing management includes dietary and nutritional support, fluid replacement, antidiarrheal treatment, and some mucosal protective therapies. Chinese clinical practice guidelines indicate that acute injury is mostly managed with graded and symptomatic treatment, and radiotherapy may be suspended in severe cases; many drugs and preventive measures still lack consistent high-quality evidence (Zhang et al., 2023). This means that sustainable, mechanism-oriented local interventions for oxidative stress and barrier injury are still lacking.

3. Efficacy and controllability of microecological interventions remain inconsistent

A systematic review included 23 RCTs with 1,919 patients; probiotics and synbiotics were associated with reduced risk of diarrhea, with risk ratios (RRs) of 0.71 and 0.45, respectively (Bartsch et al., 2021), indicating that local biological intervention in the gut has potential. However, different products vary markedly in strains, doses, and intervention durations; for engineered live bacteria, it is also necessary to simultaneously address effective retention in the target intestinal segment, boundaries of functional output, and biosafety after leaving the intended use scenario. Therefore, next-generation products should not only be “functional” but also achieve controllable local action and active exit.

2.4 Policy and Strategic Responses

The Healthy China Action-Cancer Prevention and Control Action

Implementation Plan (2023-2030) emphasizes standardized cancer diagnosis and treatment, quality control, and translation of scientific research achievements; China has also issued clinical practice guidelines for radiation-induced rectal injury to promote standardization of prevention, risk assessment, grading, and treatment of normal tissue injury. The policy direction is extending from “completing tumor treatment” further to treatment quality and full-cycle patient management.

In the field of deep space exploration, the NASA Human Research Program lists space radiation as one of the five core hazards of crewed deep-space missions. Beyond low Earth orbit, galactic cosmic rays (GCRs) and solar particle events (SPEs) constitute long-duration radiation sources; beyond physical shielding, monitoring and biomedical countermeasures are still needed.

The project’s technical concept originates from the need for radiation biological protection, but commercialization prioritizes ground-based radiotherapy-related intestinal injury, which has a clear patient base and a more mature validation pathway; after the product and clinical evidence are established, it will extend to long-term deep-space missions.

3 Pain Points of Existing Protection and Intervention Regimens

3.1 Existing Radiation Protection and Medical Intervention Regimens

Ground-based radiotherapy mainly relies on precision radiotherapy to reduce normal intestinal exposure; after injury, nutritional support, fluid replacement, antidiarrheal treatment, and other symptomatic treatments are used; microecological interventions such as probiotics and synbiotics have shown some potential. Deep-space missions mainly rely on physical shielding, mission planning, and radiation monitoring. Overall, existing solutions still lack sustained, mechanistic, and controllable intervention for local intestinal injury after radiation.

3.2 Limitations of Existing Regimens and Unmet Needs

Precision Radiotherapy

Representative solution: Intensity-Modulated Radiation Therapy (IMRT)

Core limitation: NRG/RTOG 1203 enrolled 278 patients; at the end of treatment, 33.7% of the IMRT group still had frequent or persistent diarrhea, and 7.8% used antidiarrheal medication ≥ 4 times daily (Klopp et al., 2018).

Clinical outcome: It can reduce normal tissue exposure but cannot completely avoid

acute gastrointestinal injury.

Symptomatic Antidiarrheal Treatment

Representative drug: Loperamide

Core limitation : A Phase III randomized trial enrolled 162 patients with acute radiation enteritis; after 48 h, 86.4% had diarrhea reduced to grade 0 – 1, but rebound constipation occurred in 17.3% (Abhijith et al., 2024).

Clinical outcome: It has a relatively rapid onset but mainly controls symptoms and does not directly intervene in oxidative injury or barrier recovery.

Mucosal Protective Therapy

Representative drug: Sucralfate

Core limitation : In a Phase III double-blind trial of 123 patients, the incidence of moderate-to-severe diarrhea was 53% in the sucralfate group versus 41% in the placebo group; fecal incontinence was 34% vs 16% (Martenson et al., 2000).

Clinical outcome : It did not show consistent preventive benefit, and some gastrointestinal symptoms actually increased.

Probiotics and Synbiotics

Representative solutions: Probiotics、Synbiotics

Core limitation : A meta-analysis of 23 RCTs and 1,919 patients showed reduced diarrhea risk, but strains, doses, and intervention durations varied markedly (Bartsch et al., 2021).

Clinical outcome : There is a efficacy signal, but standardization, local effective duration of action, and active safety control are insufficient.

Representative randomized study results are as follows:

**Table 1 Representative Randomized Studies of Probiotics for Preventing
Radiotherapy-Related Diarrhea**

Study	Sample Size	Probiotic	Placebo	P value
Giralt et al., 2008	85	68.2%	58.5%	0.568
Chitapanarux et al., 2010	63	9.0%	45.0%	0.002
Mego et al., 2018	54	53.8%	82.1%	<0.05
PREP, 2024	234	26.0%	22.0%	0.493

Results from different randomized studies are inconsistent, suggesting that gut microbiota intervention has potential clinical value, but its efficacy remains influenced by factors such as strain, dose, and administration regimen. Existing regimens therefore still struggle to simultaneously meet the requirements for mechanistic intervention, localized effective action, and active safety control.

4 Market Size Forecast

4.1 Global Market Size

Sybio-Gutweaver targets the prevention and adjuvant treatment of acute intestinal injury associated with abdominopelvic tumor radiotherapy. Because there are currently no independent market statistics fully corresponding to this indication, this project establishes a bottom-up market model based on core cancer patient numbers, radiotherapy demand, incidence of acute gastrointestinal toxicity, and commercialization parameters.

Table 2 Core Patient Estimates

Cancer Type	2024 Incidence	Optimal RT Utilization	Estimated RT Patients
Rectum	778,594	61%	474,942
Prostate	1,546,112	60%	927,667
Cervix Uteri	604,196	58%	350,434
Bladder	635,264	58%	368,453
Corpus Uteri	434,620	46%	199,925
Total	3,998,786	—	2,321,422

IARC GLOBOCAN 2024 shows that the above five core tumor types total approximately 4 million new cases per year globally. About 60%-80% of patients receiving abdominal and pelvic radiotherapy may develop acute gastrointestinal toxicity; extrapolated according to long-term global cancer patient growth trends, the 2025 Base Year corresponds to approximately 1.42-1.89 million people per year as the core clinical demand population.

Table 3 2025 Base Year Market Estimation

Scenario	Eligible Patients	Regional Penetration	Net Treatment Price (USD/course)	2025 Base Market (Million USD)
Conservative	1.42 M	3%-10%	\$300-1,800	97.5
Optimistic	1.89 M	5%-15%	\$700-2,400	313.8

Penetration rates and net treatment prices are assumptions of this project’s business model and are set separately according to payment capacity and market access conditions in different regions.

Table 4 Optimistic Estimate

2026E	2030E	2034E	CAGR
\$331.7 million	\$414.7 million	\$519.7 million	5.77%

Key Drivers

- 1.Global new cancer cases have reached about 20.6 million per year and are expected to increase to 34.4 million by 2050, an increase of about 67% compared with 2024.
- 2.About 60%-80% of abdominopelvic radiotherapy patients may develop acute gastrointestinal symptoms, creating sustained demand for supportive care.
- 3.Radiotherapy capacity in Asia and other emerging markets continues to improve, which will increase the number of patients who can actually receive radiotherapy and adjuvant intervention.

Table 5 Conservative Estimate

2026E	2030E	2034E	CAGR
\$100.8 million	\$115.4 million	\$132.4 million	3.46%

Key Constraints

- 1.IMRT, IGRT, and adaptive radiotherapy can reduce normal intestinal exposure but cannot yet completely avoid acute gastrointestinal toxicity.
- 2.Engineered live bacteria are a novel therapeutic modality; physician acceptance and hospital pharmacy access may limit early penetration.
- 3.Reimbursement negotiations, HTA, and regional differences in payment capacity may lower actual net treatment prices.

Risk Disclosure

1. If clinical trials cannot demonstrate improvement in key endpoints such as diarrhea severity, radiotherapy interruption rate, or quality of life, the actual applicable patient population may shrink.
2. Strain identification, potency, genetic stability, viable lyophilized bacterial count, and batch consistency constitute the main CMC risks for engineered live bacteria.
3. If actual prices, market penetration, or access speed are lower than model assumptions, the obtainable market size will be directly reduced.

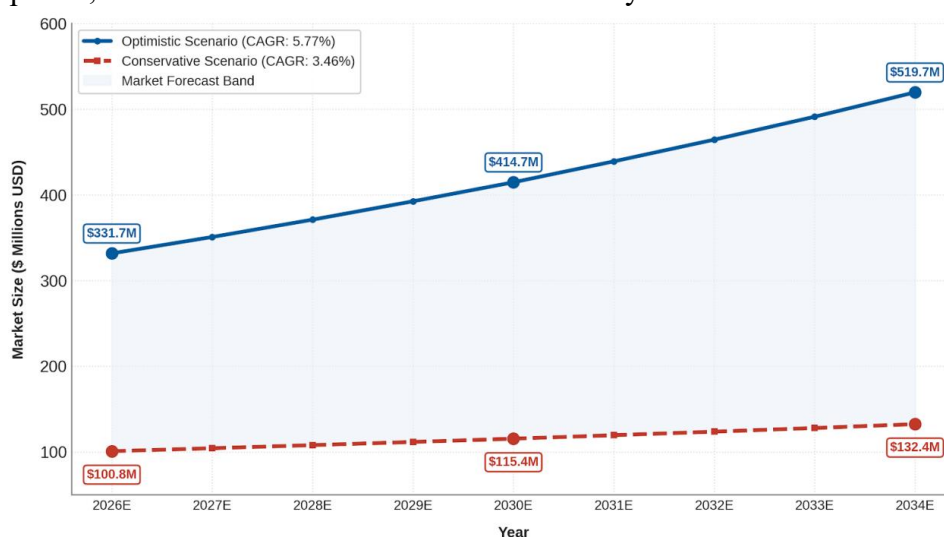


Figure 4 Global Market Growth Forecast, 2026-2034

4.2 Regional Market Size Analysis

Table 6 2025 Base Year Market Conditions

Region	2025 Scale (Million USD)	Global Share	Core Driving Factors	Main Obstacles
North America	35.3	36.2%	High payment capacity; mature RT network	LBP regulation; payer negotiation
Europe	32.4	33.2%	High RT accessibility	HTA; reimbursement pressure
Asia-Pacific	25.2	25.8%	Largest patient pool; China-first strategy	Pricing; regulatory uncertainty
Latin America	3.6	3.7%	Growing cancer burden	Limited reimbursement
Africa	1.0	1.0%	High unmet clinical need	Limited RT infrastructure
Global	97.5	100%	—	—

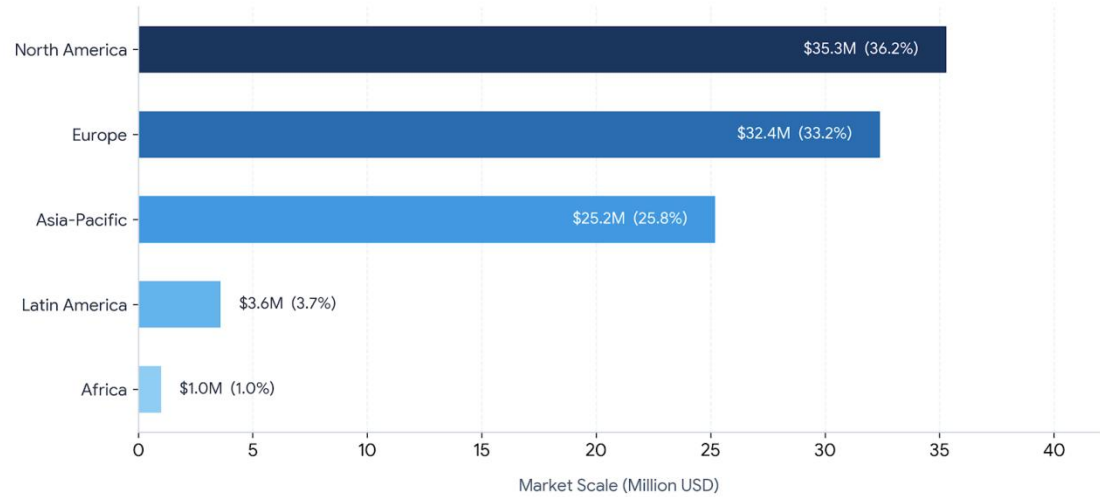


Figure 5 2025 Regional Market Distribution

The 2025 Base Year uses the conservative commercialization scenario. Asia-Pacific has the largest potential patient population, while North America and Europe account for a larger share of market value due to higher payment capacity.

Table 7 North American Market

Scenario	2026E	2030E	2034E	CAGR	Increment Logic
Conservative	36.3	40.9	46.0	3.0%	Mature market; controlled uptake

Optimistic	98.7	120.0	145.9	5.0%	Higher price; wider preventive use
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- United States: In 2026, an estimated 2,114,850 new cancer cases are expected; including 333,830 prostate cancer, 49,990 rectal cancer, 84,530 bladder cancer, 13,490 cervical cancer, and 68,270 corpus uteri cancer cases, with the five core cancer types totaling about 550,000 cases.
- Canada: In 2026, an estimated 254,100 new cancer cases and 87,900 deaths are expected; public health insurance coverage is broad, but drug price negotiations limit premiums for innovative treatments.

Table 8 European Market

Scenario	2026E	2030E	2034E	CAGR	Key Variables
Conservative	33.2	36.7	40.5	2.5%	HTA; hospital budget control
Optimistic	107.8	128.6	153.3	4.5%	High RT access; wider adoption

- EU-27: In 2024, there were about 2.7 million new cancer cases and 1.27-1.30 million cancer deaths; compared with 2022 estimates, new cases decreased by about 1.7%.
- Germany: Radiotherapy and oncology diagnosis and treatment systems are mature, but innovative therapies require strong clinical and pharmacoeconomic evidence.
- United Kingdom: NICE cost-effectiveness assessment and NHS budget constraints affect product access and actual net prices.

Table 9 Asia-Pacific Market

Scenario	2026E	2030E	2034E	CAGR	Incremental Drivers
Conservative	26.5	32.2	39.1	5.0%	Patient growth; improved RT access
Optimistic	103.3	138.0	184.3	7.5%	China-first launch; higher adoption

- China: In 2024, an estimated 5.151 million new cancer cases and about 2.58 million deaths; the large patient base makes it the first commercial market for Sybio-Gutweaver.
- Japan: In 2024, about 1.067 million new cancer cases; including about 167,000 colorectal cancer and 100,000 prostate cancer cases; the mature healthcare system supports relatively high treatment accessibility.
- India: In 2024, about 1.563 million new cancer cases, including about 79,400 cervical cancer cases; patient demand is large, but payment capacity and distribution of radiotherapy resources vary markedly.

Table 10 Latin American Market

Scenario	2026E	2030E	2034E	CAGR
Conservative	3.7	4.3	5.1	4.0%
Optimistic	16.1	20.3	25.7	6.0%

- Latin America & Caribbean: In 2024, there were about 1.581 million new cancer cases and 760,000 cancer deaths; prostate, colorectal, and cervical cancers all constitute important disease burdens.
- Brazil: One of the region's large oncology treatment markets, but public healthcare payment pressure limits net prices for innovative therapies.
- Mexico: The patient base is relatively large; private healthcare and out-of-pocket payments create some market space, but payment capacity varies markedly.

Table 11 African Market

Scenario	2026E	2030E	2034E	CAGR
Conservative	1.1	1.3	1.6	5.5%
Optimistic	5.7	7.8	10.5	8.0%

- Africa: GLOBOCAN 2024 shows about 124,600 cervical cancer cases and 111,500 prostate cancer cases, which are the main patient sources relevant to this project in the region.
- Sub-Saharan Africa: Among the 42 countries with data in the WHO survey, 16 countries have no radiotherapy equipment; insufficient radiotherapy infrastructure is a major constraint on near-term market conversion.

5 Market Analysis

5.1 PEST Analysis

5.1.1 Political Analysis

- **National cancer prevention and control policy:** The Healthy China Action-Cancer Prevention and Control Action Implementation Plan (2023-2030) was jointly issued by the National Health Commission and 13 other departments. It proposes that China's overall cancer 5-year survival rate reach 46.6% by 2030 and emphasizes standardized diagnosis and treatment, full-course management, and translation of cancer research achievements. Sybio-Gutweaver targets radiotherapy-related normal tissue injury, aligning with the policy direction of cancer treatment shifting from "prolonging survival" to "placing equal emphasis on efficacy and quality of life".
- **National strategic support for synthetic biology:**
 - The 14th Five-Year Plan for the Development of the Bioeconomy lists synthetic

biology, new drug creation, and biosafety as frontier priority areas;

-It explicitly proposes breakthroughs in key technologies such as computational strain design, high-throughput screening, efficient expression, and precise regulation;

-It promotes the application of synthetic biology in new drug development and disease treatment, providing a policy foundation for the technological translation of engineered EcN.

- **Regulatory framework for live bacterial drugs is gradually being established:**

-The 2025 edition of the Chinese Pharmacopoeia was implemented on October 1, 2025. Volume III continues to include the General Chapter on Microecological Live Bacterial Products, and viable count, stability, and strain quality control already have a national drug standard basis;

-The U.S. FDA has issued guidance specifically for early clinical CMC for INDs of Live Biotherapeutic Products (LBPs), setting clear requirements for strain identification, manufacturing, quality, and stability, providing an international regulatory interface for the project's "NMPA primary pathway, FDA parallel assessment" strategy.

- **Gansu industrial policy support:**

-During the 14th Five-Year Plan period, the output value of Gansu's biopharmaceutical industry increased by 83.5%, and the industrial base continued to expand;

-The latest industrial policy proposes forming a RMB 50 billion-level biopharmaceutical industry by 2028;

-The 15th Five-Year Plan further focuses on vaccines, antibody drugs, recombinant proteins, and cell and gene therapies, and promotes the miniaturization and scaled development of medical heavy ion accelerators.

5.1.2 Economic Analysis

- Accelerated commercialization of live bacterial therapeutics: As microbial therapeutic products enter clinical use and the market, industry value has extended from technical validation to standardized manufacturing, oral delivery, and license-out. Sybio-Gutweaver uses EcN fermentation, lyophilized live bacteria, and gut-release capsules, and reduces late-stage clinical and industrialization investment through co-development/technology licensing.

Table 12 Representative Commercial Cases of Live Bacterial/Microbial Therapeutics

Enterprise / Product	Core Product	Commercial Performance
Seres / VOWST	FDA-approved oral	Net sales of USD 19.6

	microbial product	million in about 7 months after launch
Ferring / REBYOTA	Live microbiota product	2026 WAC approximately USD 9,411 per dose
Seres + Nestlé	VOWST commercial collaboration	upfront + USD 125 million milestones
SER-301	Oral fermented microbial product	Phase Ib initiation triggered USD 10 million milestone

The economic advantages of live bacterial therapeutics are mainly reflected in three aspects:

-Fermentation manufacturing: microbial expansion is suitable for scaled manufacturing;

-Oral delivery: reduces invasive administration and occupation of medical resources;

-Asset-light model: transfers late-stage clinical and commercialization costs through milestone payments and license-out.

- Payment environment improvement:

- The 2025 medical insurance catalog added 114 drugs, 50 of which are Class 1 innovative drugs;

- The first edition of the commercial insurance innovative drug list included 19 drugs, further broadening payment channels for innovative drugs.

5.1.3 Social Analysis

- Large cancer patient base:

- China has about 4.825 million new cancer cases each year;

- About 2.574 million cancer deaths;

- About 10.969 million 5-year prevalent cases.

- The expanding long-term cancer survivor population continues to increase the importance of treatment-related side effects and full-cycle patient management.

- Population aging continues to increase demand: In 2025, China's population aged 60 and above reached 323 million, accounting for 23.0%; the population aged 65 and above reached 224 million, accounting for 15.9%. As the population at high risk of cancer increases, demand for radiotherapy and normal tissue protection is expected to persist long term.

- There remains a significant unmet need for radiotherapy-related intestinal injury:

- About 60%-80% of patients receiving abdominal and pelvic radiotherapy may develop acute gastrointestinal toxicity;

- Even with IMRT, 33.7% of patients in NRG/RTOG 1203 still had frequent or persistent diarrhea at the end of treatment;
- 7.8% of patients needed to use antidiarrheal medication ≥ 4 times daily.
- Intestinal biological intervention already has a clinical acceptance basis: A systematic review of 23 RCTs and 1,919 patients showed that probiotics were associated with reduced diarrhea risk (RR=0.71), and synbiotics RR=0.45. This indicates that there is real clinical demand for local intestinal biological intervention and also leaves room for upgrading to more controllable engineered live bacteria.

5.1.4 Technological Analysis

Core advantages:

Functional Module	Innovation	Industrial Value
DHICA-optimized melanin	Alleviate ROS and oxidative injury	Target upstream mechanisms of radiation injury
Lactate module	Support intestinal epithelial barrier recovery	Extend from symptom control to tissue repair
Ag43 controllable adhesion	Enhance short-term retention in target intestinal segment	Enhance local effective exposure
Dual safety switches	Qwitch + L-arabinose/hok-sok	Reduce risk of abnormal persistent output and escape
Lyophilized gut release	Storage—acid resistance—intestinal release	Improve usability of oral products

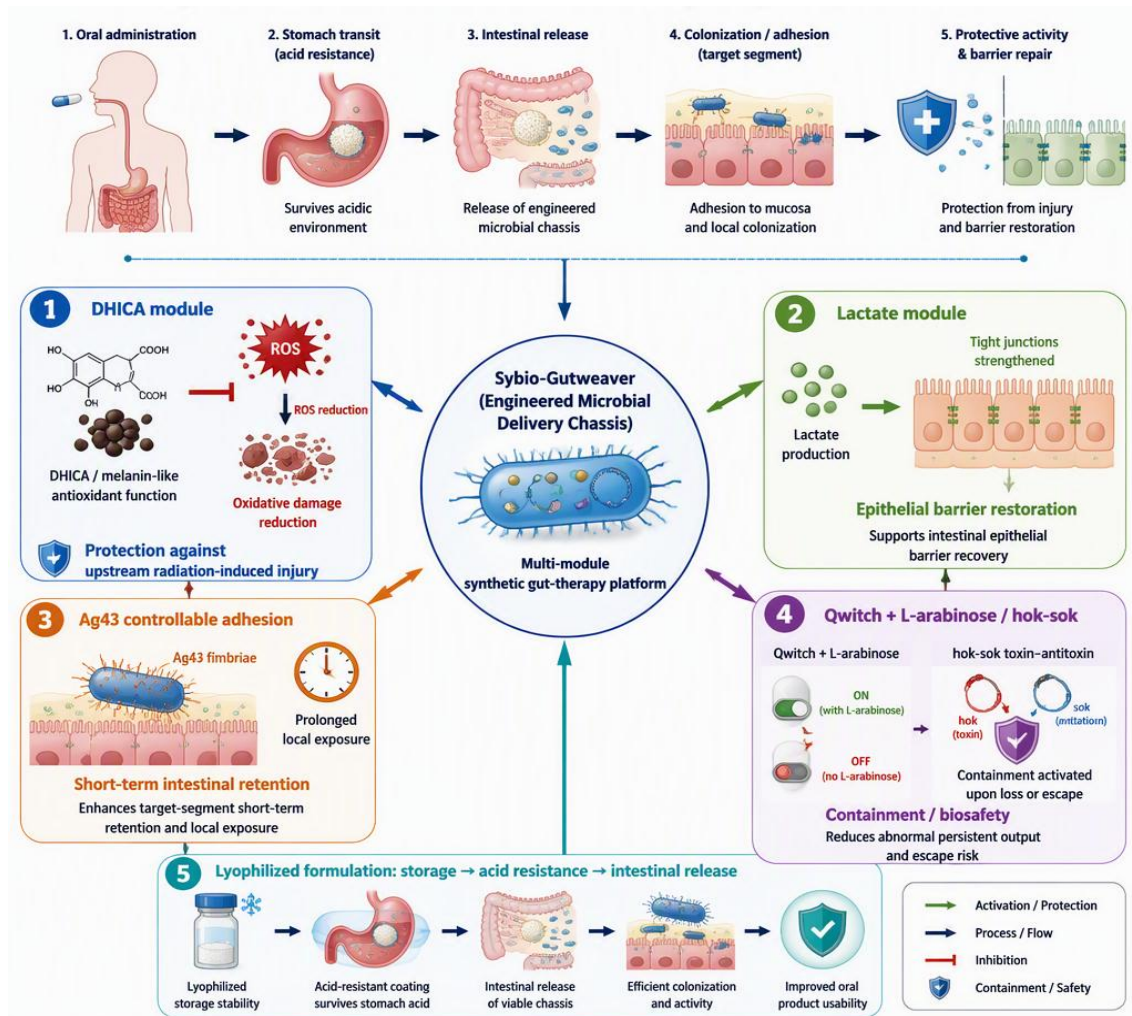


Figure 6 Multi-module synergistic mechanism of Sybio-Gutweaver

Compared with traditional probiotics, Sybio-Gutweaver integrates antioxidant activity, barrier support, local retention, and active exit into the same EcN chassis, and restricts unintended persistent colonization through dual biosafety mechanisms.

- Oral live bacterial formulations already have clinical validation precedents: In a randomized controlled study, the FDA-approved oral live bacterial product VOWST reduced the recurrence rate of *Clostridioides difficile* infection within 8 weeks from 39.8% in the placebo group to 12.4%, indicating that standardized oral live microbial products can achieve clear clinical efficacy endpoints.
- Engineered bacteria still carry translation risks: Synlogic's engineered bacterial drug SYN1934 entered a pivotal Phase III trial but was terminated in 2024 because interim data suggested difficulty in achieving the primary efficacy endpoint, and the termination was not due to safety issues. This indicates that the core challenge of commercializing engineered bacteria has shifted from “whether it can be constructed” to “whether it can produce sufficiently large clinical benefit.”

5.2 SWOT Analysis

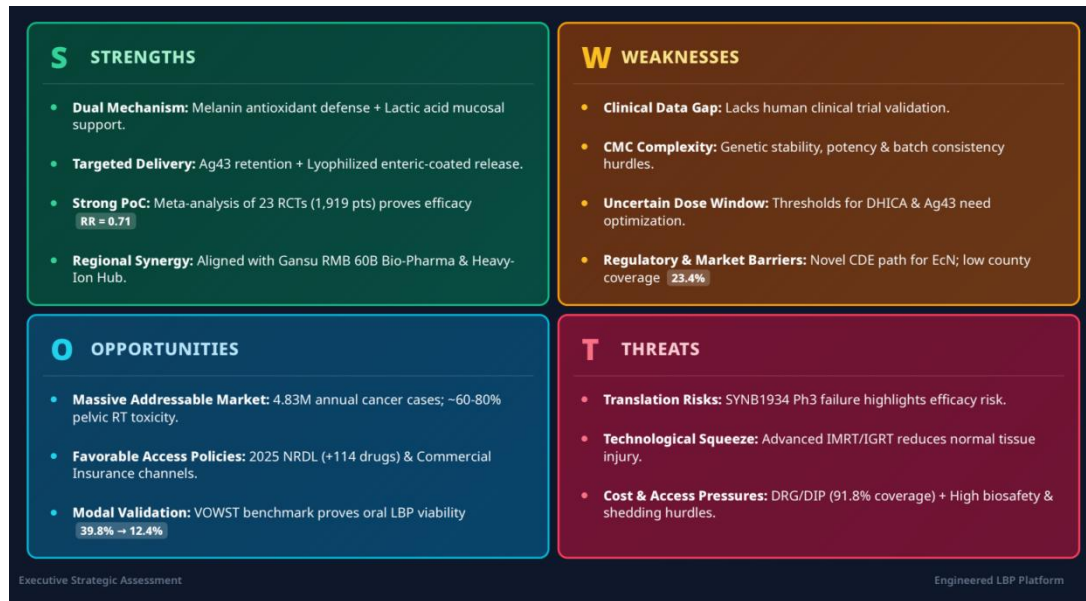


Figure 7 Sybio-Gutweaver SWOT Analysis

5.2.1 Strengths

Internal:

- 5 productization modules + 2 layers of active safety control
- Melanin antioxidant activity and lactate barrier support form dual-mechanism intervention
- Ag43 controllable short-term retention + lyophilized intestinal release improve local effective action

External:

- 23 RCTs and 1,919 patients have demonstrated a clinical efficacy signal for intestinal biological intervention (Probiotics RR=0.71)
- Gansu's biopharmaceutical industry has a 2030 target of RMB 60 billion and is developing the heavy ion therapy industry

5.2.2 Weaknesses

Internal:

- Currently lacks human clinical data
- Multi-functional circuits increase the difficulty of controlling genetic stability, potency, and batch consistency
- Effective doses and safety thresholds of DHICA, lactate, and Ag43 still need further convergence

External:

- Domestic registration experience for engineered EcN is limited, and

product-specific requirements need to be confirmed with CDE on a stage-by-stage basis

- There are only 59 county-level radiotherapy units in western China; the radiotherapy coverage rate in counties with a population of more than 500,000 is only 23.4%, limiting early grassroots market penetration

5.2.3 Opportunities

- China has about 4.825 million new cancer cases annually and about 10.969 million 5-year prevalent cases
- About 60%-80% of abdominopelvic radiotherapy patients develop acute gastrointestinal toxicity
- The 2025 medical insurance catalog added 114 drugs, 50 of which are Class 1 innovative drugs
- The first commercial insurance innovative drug list included 19 innovative drugs, forming a supplementary payment channel
- VOWST reduced recurrence rate from 39.8% to 12.4%, validating the productization potential of oral live bacterial drugs

5.2.4 Threats

- Engineered bacterium SYNB1934's Phase III was terminated due to insufficient efficacy endpoints, highlighting clinical translation risk
- IMRT, IGRT, and adaptive radiotherapy continue to reduce normal tissue exposure, potentially compressing some preventive demand
- The coverage rate of payment by disease category reached 91.8%, increasing hospital budget constraints and medical insurance payment pressure.
- Engineered live bacteria involve strain escape, genetic stability, and environmental biosafety, with regulatory thresholds higher than ordinary probiotics
- The speed of physician acceptance of novel engineered live bacterial therapies may affect early hospital access

6 Commercialization Path

6.1 Technology and Intellectual Property Protection

The commercial value of Sybio-Gutweaver is built on core technologies such as engineered strains, functional circuits, biosafety control, and delivery systems. The project intends to form protective barriers through three aspects: intellectual property, technology confidentiality, and rapid translation.

6.1.1 Intellectual Property

The team has submitted patent applications around the core technology of Sybio-Gutweaver, focusing on protecting engineered strain construction, functional module integration, and their application in intervention for radiotherapy-related intestinal injury.

On the basis of existing applications, the team plans to further improve the patent layout around key technologies such as antioxidant activity, barrier support, controllable adhesion, dual biosafety control, and intestinal delivery, and to advance subsequent domestic and international intellectual property protection according to the commercialization process, establishing core technical barriers for co-development and technology licensing.

6.1.2 Confidentiality Measures

Patents cannot cover all technical details of the project. For content not yet patented or not suitable for full disclosure, the project will use trade secret protection, especially technical information such as strain construction parameters, key culture conditions, functional module regulation thresholds, lyophilization and recovery conditions, and quality control methods.

During internal R&D, graded management is implemented for strain sequences, plasmid maps, experimental data, and key processes; in external exchanges, technical information is disclosed gradually according to the cooperation stage. For in-depth exchanges involving investors, research institutions, pharmaceutical companies, and manufacturing partners, a non-disclosure agreement (NDA) should be signed before disclosing core technology.

After entering the sample validation or co-development stage, material transfer agreements (MTAs) and co-development agreements will further clarify sample use, data use scope, background intellectual property, and ownership of intellectual property generated through cooperation, thereby reducing the risk of core technology leakage and ownership disputes during cooperation.

6.1.3 Rapid Market Conversion

Biomedical projects have long R&D cycles, and the actual commercial value of patents is affected by clinical development and registration cycles. Therefore, on the basis of ensuring safety and compliance, this project needs to shorten as much as possible the time from laboratory validation to candidate product selection.

At the current stage, the team will prioritize completing core strain convergence, functional validation, biosafety switches, genetic stability, and optimization of the lyophilization-intestinal release system; subsequently, it will gradually enter animal

efficacy and safety evaluation, manufacturing process development, quality standard establishment, and registration preparation.

For capital- and resource-intensive stages such as late-stage clinical trials, scaled manufacturing, and commercial promotion, the project does not plan to undertake all of them independently. Instead, it will engage in advance with companies experienced in live bacterial drugs, biopharmaceuticals, or microecological product development and carry out co-development or technology licensing after key data mature.

This model enables the team to continue concentrating resources on core engineered bacteria and the technology platform while leveraging industrial partners' capabilities in clinical, manufacturing, registration, and sales links to improve overall translation efficiency.

6.1.4 Disclaimer and Phased Statements

The team has submitted patent applications for the core technology of Sybio-Gutweaver, but as of the formation stage of this business plan, the relevant applications have not yet been granted. Given that patent examination takes a certain period, the current scope of intellectual property protection and the final authorization results are still subject to review by the relevant competent authorities.

In competition presentations and external exchanges, the team will control the disclosure scope of undisclosed core technologies and continue to advance supplementary patent applications based on subsequent R&D results. For in-depth technical exchanges involving investors, partners, and other external entities, the use and dissemination of sensitive information will be restricted through non-disclosure agreements (NDAs) and other means to reduce the risk of intellectual property leakage.

6.2 Value Chain Analysis

From laboratory concept to final commercialization, engineered live bacterial drugs generally need to go through multiple stages, including strain design and screening, functional validation, formulation development, preclinical research, clinical development, regulatory submission, scaled manufacturing, and market sales.

Sybio-Gutweaver is currently mainly in the drug discovery and early product development stage. The team has carried out systematic design around melanin antioxidant activity, lactate barrier support, Ag43 adhesion, and dual biosafety mechanisms, and has further considered productization issues such as lyophilization, gastric acid protection, and intestinal release.

At the front end of the value chain, the team's advantages are concentrated in synthetic biology design, engineered bacterial construction, metabolic modeling, and functional module development; as the product enters later stages, demand for GLP

safety evaluation, clinical trials, GMP manufacturing, quality systems, and drug registration capabilities will increase rapidly.

Therefore, this project intends to concentrate R&D resources on core strains, functional mechanisms, safety control, and candidate product selection, and supplement subsequent resources through research institutions, hospitals, CROs/CDMOs, and pharmaceutical companies. After relatively complete preclinical evidence is available, part of the clinical development, scaled manufacturing, and market promotion work will be undertaken by industrial partners through co-development or license-out.

This value chain division of labor can reduce the financial pressure on the team from independently undertaking full-chain drug development while preserving the long-term value of core intellectual property and the technology platform.

6.3 Commercialization Strategy

Based on the value chain analysis, Sybio-Gutweaver will adopt an asset-light model of “independent R&D + co-development/technology licensing”. The team will maintain R&D advantages in core strains, functional circuits, and intellectual property, and after key technical and clinical milestones, introduce industrial partners to undertake subsequent manufacturing, registration, and market promotion.

6.3.1 Commercialization Strategy Environment

Through an interview with Gao Rencheng, CEO of Jiyin Chuangwu (Shanghai) Technology Co., Ltd., we found that Sybio-Gutweaver is currently in the early stage of translating laboratory technology into a candidate product. The project adopts a phased commercialization strategy of “ground-based radiotherapy first, long-term extension to space scenarios”. At the current stage, priority is given to completing product validation, clinical development, and market access around abdominopelvic radiotherapy-related intestinal injury. The team has currently formed a multi-functional engineered EcN live bacterial scheme, but key complementary assets such as scaled manufacturing, clinical development, and regulatory submission still need to be supplemented.

- Insufficient scaled manufacturing capacity: lyophilized live bacteria, intestinal release, genetic stability, and batch consistency still require further validation and process scale-up.
- Clinical data are still insufficient: efficacy, safety, and applicable population still need to be validated through animal experiments and subsequent clinical studies.
- Limited registration and market access capabilities: live engineered bacteria involve drug registration, quality control, and biosafety requirements, requiring professional industrial resource support.

Therefore, near-term resources will focus on ground-based radiotherapy product development, and external cooperation will supplement manufacturing, clinical, and registration capabilities; at the current stage, the space direction will mainly focus on scientific validation and technology reserves to avoid advancing two high-cost commercial routes simultaneously.

6.3.2 Partners and Resources

Through research and exchanges with research institutions, biopharmaceutical companies, and related units in the industrial chain, the team has gradually formed a multi-level industrial resource network covering R&D, technology platforms, manufacturing translation, and market applications around the R&D, manufacturing, quality control, and achievement translation needs of Sybio-Gutweaver.



Figure 8 Partners

Existing cooperation and industrial resources cover key links in project commercialization, mainly including:

R&D stage: School of Stomatology, Lanzhou University; Gansu Provincial Hospital; Gansu Wuwei Tumor Hospital; Institute of Modern Physics, Chinese Academy of Sciences, etc., providing scientific research and clinical resource support for radiation injury mechanisms, clinical needs, biosafety, and product evaluation.

Upstream technology platforms: Lanzhou Baiyuan Gene Technology Co., Ltd.; Lanzhou Huazhi Tiancheng Biotechnology Co., Ltd., etc., covering biochips, molecular diagnostics, functional strains, and biotechnology R&D platforms.

Midstream R&D and manufacturing: Lanzhou Institute of Biological Products Co., Ltd.; Gansu Haotian Technology Co., Ltd.; Lanzhou Foci Pharmaceutical Co., Ltd., etc., covering biologics R&D, drug processes, GMP quality systems, CDMO/CMO, formulation manufacturing, and scaled translation.

Downstream applications and products: Gansu Kangsheng Pharmaceutical and Health Products Group Co., Ltd., etc., providing industry-side references for

pharmaceutical product distribution, warehousing and delivery, quality traceability, market access, and achievement translation.

The team will make full use of the above scientific research and industrial resources to strengthen exchanges and cooperation in experimental validation, process development, quality control, clinical evaluation, and industrial translation in subsequent R&D, providing resource support for advancing Sybio-Gutweaver from laboratory R&D to productization and commercialization.

6.3.3 Commercialization Advancement Steps

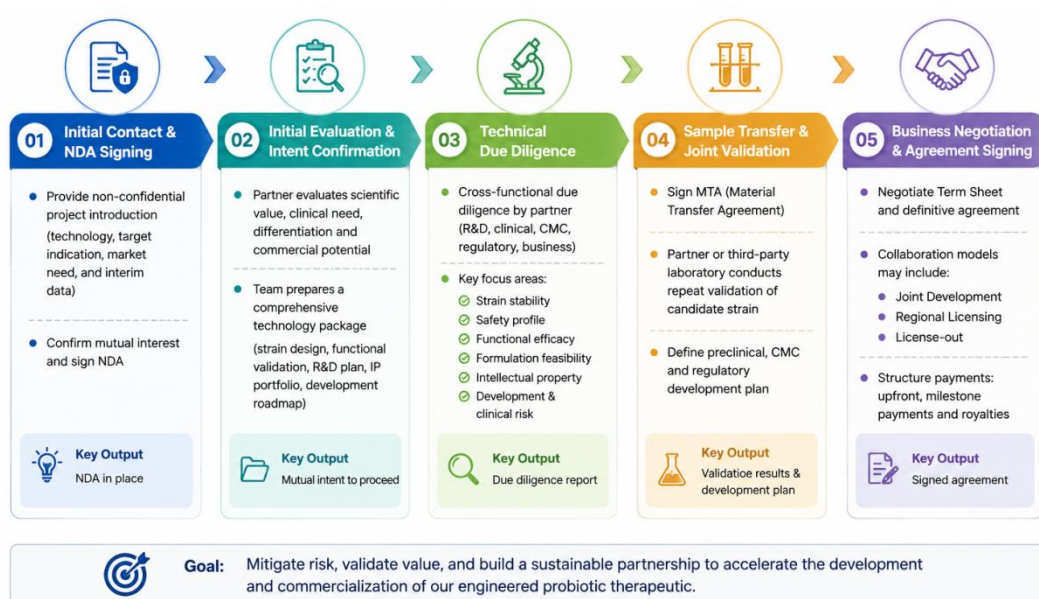


Figure 9 Commercialization Steps

1. Initial contact and NDA signing

The team first provides potential partner companies with a non-confidential project introduction, including technical principles, target indications, market demand, and staged experimental results. After both parties confirm intention for further exchange, a non-disclosure agreement is signed before entering the core technology information disclosure stage.

2. Preliminary evaluation and confirmation of cooperation intention

Partner companies conduct preliminary evaluation from aspects such as scientific value, clinical demand, product differentiation, and commercial potential. The team simultaneously prepares a relatively complete technical information package, including strain design, functional validation, R&D plan, intellectual property, and product development roadmap.

3. Technical due diligence

After entering the in-depth cooperation stage, the partner's R&D, clinical, manufacturing, regulatory, and business teams jointly conduct due diligence, focusing on evaluating engineered bacterial stability, safety, functional efficacy, formulation

feasibility, intellectual property, and subsequent clinical development risks.

4. Sample transfer and joint validation

After passing preliminary due diligence, both parties sign an MTA, and the partner or a third-party laboratory performs repeated validation of candidate strains and further determines preclinical, CMC, and regulatory development plans.

5. Business negotiation and agreement signing

After completing key validation, both parties enter Term Sheet and formal agreement negotiations. Cooperation forms may include co-development, regional licensing, or license-out, with upfront payments, milestone payments, and sales revenue sharing set according to the development stage.

7 Development Strategy

7.1 STP Analysis—Market Segmentation

Different countries and regions differ significantly in cancer patient scale, radiotherapy resources, regulatory requirements for live bacterial drugs, healthcare payment capacity, and industrial ecosystem. Based on clinical needs, product development feasibility, market access conditions, and long-term commercial value, Sybio-Gutweaver divides its target market into three regions: Asia-Pacific, North America, and Europe.

The project originates from the need for deep-space radiation health protection, but near-term commercialization should prioritize ground-based medical scenarios with a clear patient base, quantifiable clinical endpoints, and a more mature validation pathway. Therefore, the project adopts a development strategy of “ground-based radiotherapy first, parallel preparation for international markets, and long-term extension to space medicine,” with radiotherapy-related acute intestinal injury in abdominal and pelvic tumors as the near-term entry point.

7.1.1 Asia-Pacific Market

The Asia-Pacific region has a large cancer patient base and continuously growing radiotherapy demand, but countries and regions differ markedly in healthcare payment capacity, radiotherapy accessibility, and innovative drug access conditions. The project’s conservative commercialization model shows that the Asia-Pacific market size in 2025 is USD 25.2 million, accounting for 25.8% of the global target market; it is expected to grow to USD 26.5 million in 2026.

China: lead market. The National Cancer Center of China estimates that in 2022 China had about 4.8247 million new cancer cases, and demand for cancer treatment and management of treatment-related side effects persists (Han et al., 2024). The

project intends to advance along the drug R&D pathway and, relying on the communication mechanism of the Center for Drug Evaluation (CDE) of the National Medical Products Administration, communicate on product attributes, preclinical research, CMC, genetic stability, biosafety, and clinical development plans during stages such as clinical trial applications, key clinical studies, and marketing authorization applications (CDE, 2026).

The focus of the Chinese market is to rely on existing hospitals, heavy ion centers, and research collaboration resources to prioritize candidate product convergence, clinical demand validation, and early clinical development, and then gradually enter major cancer hospitals and radiotherapy centers nationwide.

Japan and South Korea: high-accessibility extension markets. Japan and South Korea have mature healthcare systems and a high degree of standardization in oncology diagnosis and treatment, and also have high requirements for the quality, clinical evidence, and payment value of innovative therapies. After the project generates safety and preliminary efficacy evidence in China, these two countries can be considered as subsequent expansion directions in high-quality Asia-Pacific healthcare markets.

India and other emerging markets: long-term reserve markets. Such markets have relatively large patient bases, but price accessibility, supply chains, cold-chain storage and transportation, and distribution of radiotherapy resources still require separate assessment, and they are not currently prioritized for early entry.

7.1.2 North American Market

The North American market is centered on the United States and has a mature oncology diagnosis and treatment network, relatively high payment capacity for innovative therapies, and an active microbial therapeutics industrial ecosystem. The American Cancer Society estimates that in 2026 the United States will have about 2.1149 million new cancer cases and about 626,100 cancer deaths, providing a stable long-term demand basis for cancer supportive care (ACS, 2026).

In June 2016, the U.S. FDA issued Early Clinical Trials With Live Biotherapeutic Products: Chemistry, Manufacturing, and Control Information: Guidance for Industry, providing CMC guidance for IND submissions for early clinical trials of live biotherapeutic products, focusing on strain source and identification, manufacturing process, quality control, stability, and consistency of clinical trial products (FDA, 2016). This guidance focuses on CMC information for early LBP development and is not a complete clinical development regulatory system.

Therefore, the near-term value of the North American market mainly lies in international validation and industrial cooperation. While conducting preclinical and early clinical validation in China, the project will prepare CMC materials in parallel,

including strain banks, genetic stability, viable bacterial potency, formulation release, biosafety switches, batch consistency, and environmental risk control; after a reliable data package is formed, it will evaluate the feasibility of co-development, regional licensing, or license-out. Because technologies such as IMRT, IGRT, and adaptive radiotherapy continue to reduce normal intestinal radiation doses, the candidate product still needs to demonstrate incremental value in endpoints such as acute gastrointestinal toxicity, radiotherapy completion, quality of life, and use of supportive care resources.

7.1.3 European Market

The European market has a well-developed radiotherapy system, a high level of clinical standardization, and a mature health technology assessment mechanism. In 2022, the European Union had about 2.7 million new cancer cases and about 1.3 million cancer deaths, with long-term demand for cancer supportive care (ECIS, 2023).

Continental EU: evidence-based access market. Western European countries such as Germany and France have relatively strong payment capacity for innovative drugs and relatively mature demand for cancer supportive care, and can serve as priority areas for the project's future entry into continental Europe. However, EU countries differ markedly in HTA, medical insurance payment, and price negotiation mechanisms, and the product needs sufficient clinical and pharmacoeconomic evidence.

In 2016, the EMA launched the PRIME scheme to provide early regulatory support for innovative drugs that may address unmet clinical needs and offer significant advantages over existing treatments. Research institutions and small and medium-sized enterprises may apply for Early Entry PRIME support when they have convincing nonclinical proof of concept and first-in-human data (EMA, 2016). The project will evaluate the applicability of this pathway after obtaining early human safety and preliminary efficacy data, but will not include it in near-term commercialization forecasts.

United Kingdom: independent assessment market. The United Kingdom needs separate assessment according to its own regulatory, payment, and hospital access processes. The project will not separately invest commercialization resources in the early stage, and will advance only after the continental EU market access strategy and clinical value evidence mature.

After completing geographic and patient-scenario segmentation, the project also needs to further identify partner centers with clinical research and standardized radiotherapy capabilities. The screening criteria for the first batch of centers are shown in the table below.

Table 13 Screening Criteria for First-Batch Clinical Partner Centers

Screening Dimension	Industry Background Data	Project Access Criteria	Strategic Significance
Radiotherapy service basis	In 2018, mainland China had 1,463 radiotherapy units, with about 1.2596 million radiotherapy patient visits annually	Prioritize centers that stably perform abdominopelvic tumor radiotherapy and have continuous patient sources	Ensure target patient accessibility and improve enrollment efficiency
Equipment and technology basis	Nationwide, there were 2,021 linear accelerators and 355 CT simulators	Have CT simulation positioning, modern external beam treatment planning, and dose verification capabilities	Improve consistency of treatment processes and data collection
Personnel and quality control capability	Nationwide, there were 29,096 radiotherapy practitioners, including 4,172 medical physicists	Have full-time medical physicists; have radiotherapy quality control and adverse event reporting processes	Facilitate establishment of safety and efficacy evidence

7.2 STP Analysis—Target Market Selection

Considering patient scale, clinical resources, regulatory feasibility, and market entry conditions, the project adopts the target market strategy of “China ground-based radiotherapy as the lead, North America in parallel, Europe for subsequent expansion, and space medicine as long-term extension.”

China is the project’s primary target market. In 2018, mainland China had 1,463 radiotherapy units and about 1.2596 million radiotherapy visits annually, forming a relatively broad radiotherapy service and clinical collaboration network (Zhang Ye et al., 2020). Combined with the team’s existing hospital, heavy ion center, and research collaboration resources, the project can prioritize candidate product validation, preclinical research, and early clinical development in China to generate evidence on safety, delivery stability, and clinical value.

The United States is the project’s secondary target market. Its core value lies not in near-term independent sales but in international CMC standards, a mature innovative therapy evaluation system, and industrial cooperation value. The project will focus on intellectual property layout, construction of an international development interface, and preparation of CMC materials for early LBP development, creating conditions for future co-development or regional licensing.

Europe is the project’s subsequent target market. Europe has stable demand for cancer supportive care and relatively high payment capacity, but market entry relies more on high-quality clinical evidence, HTA assessment, and pharmacoeconomic materials. The project will continuously track European regulatory and payment conditions and gradually expand after product value evidence matures.

The space medicine scenario is a long-term strategic extension and is not included in near-term revenue forecasts or commercialization assessments.

Deep-space radiation protection has a limited applicable population, a long validation cycle, and is highly dependent on the mission environment. The project will first validate the safety, local action, and biological protection potential of the engineered bacteria in ground-based radiotherapy scenarios, and then evaluate its application value in space medicine.

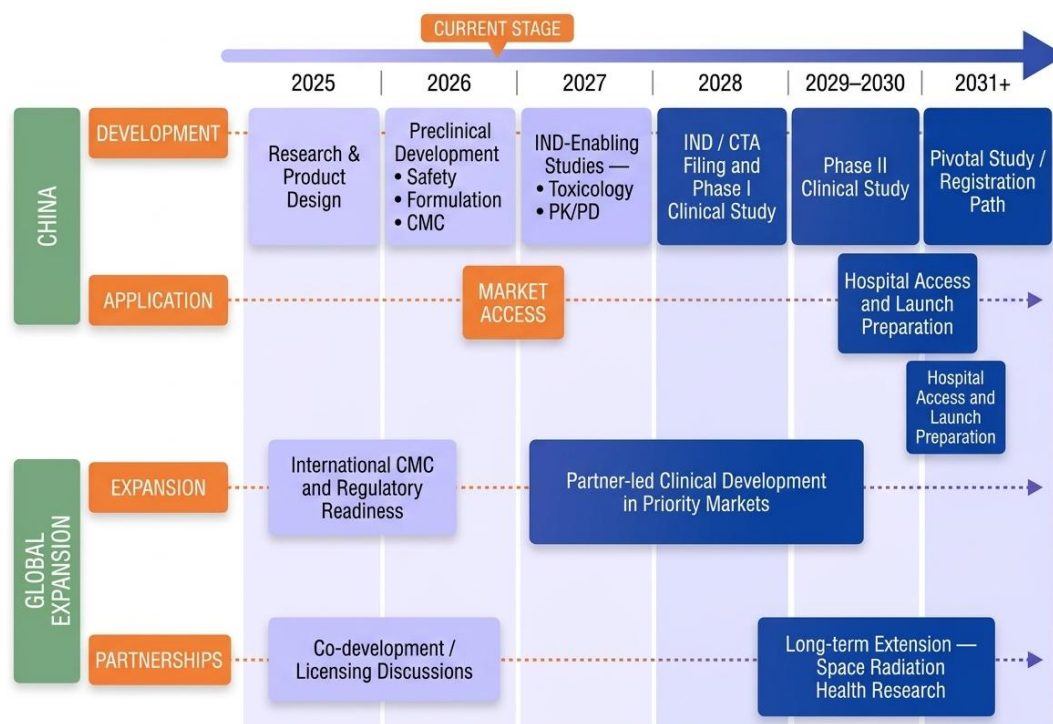


Figure 10 Sybio-Gutweaver Market Entry and Development Roadmap

7.3 STP Analysis—Market Positioning

Sybio-Gutweaver is positioned as an oral engineered live bacterial candidate formulation for radiotherapy-related acute intestinal injury, providing a mechanism-oriented supplementary option to existing symptomatic supportive care through antioxidant alleviation, intestinal barrier support, short-term local retention, and active safety exit.

This positioning addresses the persistent demand for gastrointestinal toxicity after radiotherapy. The NRG/RT0G 1203 study showed that among patients with cervical and endometrial cancer, even after receiving pelvic IMRT, 33.7% still reported frequent or nearly persistent diarrhea at the end of radiotherapy, and 7.8% used antidiarrheal medication no fewer than 4 times daily (Klopp et al., 2018). This indicates that although precision radiotherapy can reduce gastrointestinal toxicity, it cannot yet completely eliminate radiotherapy-related intestinal injury.

Our project shifted the technical focus from the initial hypothesis of “radiation dose attenuation” to oxidative stress alleviation, intestinal barrier support, and biosafety control: the melanin module is used to explore ROS alleviation, the lactate

module to support the intestinal epithelial barrier, the Ag43 module to enhance short-term retention of engineered bacteria in the target intestinal segment, and dual safety switches to limit unintended persistent output and dissemination risk. Therefore, Sybio-Gutweaver is not positioned as an ordinary probiotic, physical shielding material, or substitute for precision radiotherapy, but as a candidate therapeutic regimen featuring local mechanistic intervention and active safety control. Prioritizing ground-based radiotherapy will also accumulate functional and safety evidence for its future space medicine applications.

7.4 Marketing

7.4.1 Marketing Objectives

The project intends to develop along the drug pathway, with early market work centered on scientific communication, physician awareness building, clinical evidence communication, and hospital access preparation. The project will establish professional awareness in the fields of radiotherapy-related intestinal injury and engineered live bacterial therapy, clarifying clinical needs, technical logic, and safety boundaries; enhance physicians' awareness and acceptance of engineered live bacterial candidate therapies through expert interviews, investigator exchanges, and academic conferences; and gradually build a collaboration network among radiotherapy, oncology, gastroenterology, pharmacy departments, and clinical research institutions to provide a basis for clinical research and real-world evidence accumulation.

In the later stage of product development, the project will develop medical education, pharmacoeconomic, and safety communication materials for medical institutions to support hospital pharmacy access and market access assessment. After product approval and launch, under the guidance of medical affairs and medical institutions, it will carry out public education on radiotherapy-related intestinal injury, standardized medication, and safe use to improve patients' understanding of and adherence to treatment regimens.

7.4.2 Marketing Strategies

We recently conducted online interviews with two representative marketing practitioners—Xiaohongshu blogger Louise and Xiaohongshu marketing service provider Raccoon Commune. Louise has long focused on brand strategy, marketing department performance, and brand growth, and has worked in brand strategy and marketing for hundred-billion-level brands, with perspectives from both brand owners and managers; Raccoon Commune mainly provides enterprises with Xiaohongshu account operation, traffic growth, and marketing implementation services, and has

relatively rich platform operation experience. Through the two interviews, the team further understood the basic logic of Xiaohongshu marketing from content planning and account operation to traffic conversion from two levels: brand strategy and platform execution. Specifically, brand marketing mainly includes the following four aspects:

- 1.Brand positioning: clarify target audience, brand value proposition, and core communication messages, establish a unified external expression, avoid fragmentation of content across multiple channels, and gradually build stable brand awareness.

- 2.Content operation: naturally integrate brand information into content that users follow through knowledge sharing, scenario expression, and viewpoint output. Compared with simple product promotion, content operation emphasizes the match between brand value and user needs to enhance user awareness and trust.

- 3.Community/circle communication: select suitable bloggers, industry experts, and opinion leaders based on target audience profiles, and achieve circle diffusion through interviews, content co-creation, and thematic sharing. The selection of partners focuses on audience match, content quality, and interaction performance, rather than simply referring to follower size.

- 4.Effect evaluation: set brand building, user reach, and actual conversion goals according to different marketing stages, comprehensively examine brand exposure, content interaction, user search, cooperation leads, and conversion effects, and adjust content direction and resource investment in a timely manner through data review.

In terms of marketing service investment, brands can choose different service combinations according to their development stage. Account operation mainly covers daily updates, content planning, and data analysis; when broader dissemination is needed, content co-creation, influencer cooperation, and platform promotion can be further combined to form a relatively complete marketing plan. Specific service content and investment scale should be adjusted according to brand positioning, target audience, communication cycle, and marketing objectives to achieve rational resource allocation.

After internal evaluation and discussion, before the launch of clinical trials, the team will temporarily not purchase large-scale third-party marketing services and will prioritize using self-owned platforms such as WeChat official accounts and Xiaohongshu to complete brand positioning and content operation. When the project advances to the clinical trial stage, professional marketing agencies and vertical-domain content creators will be introduced according to actual market demand to further amplify brand influence and improve market conversion efficiency.

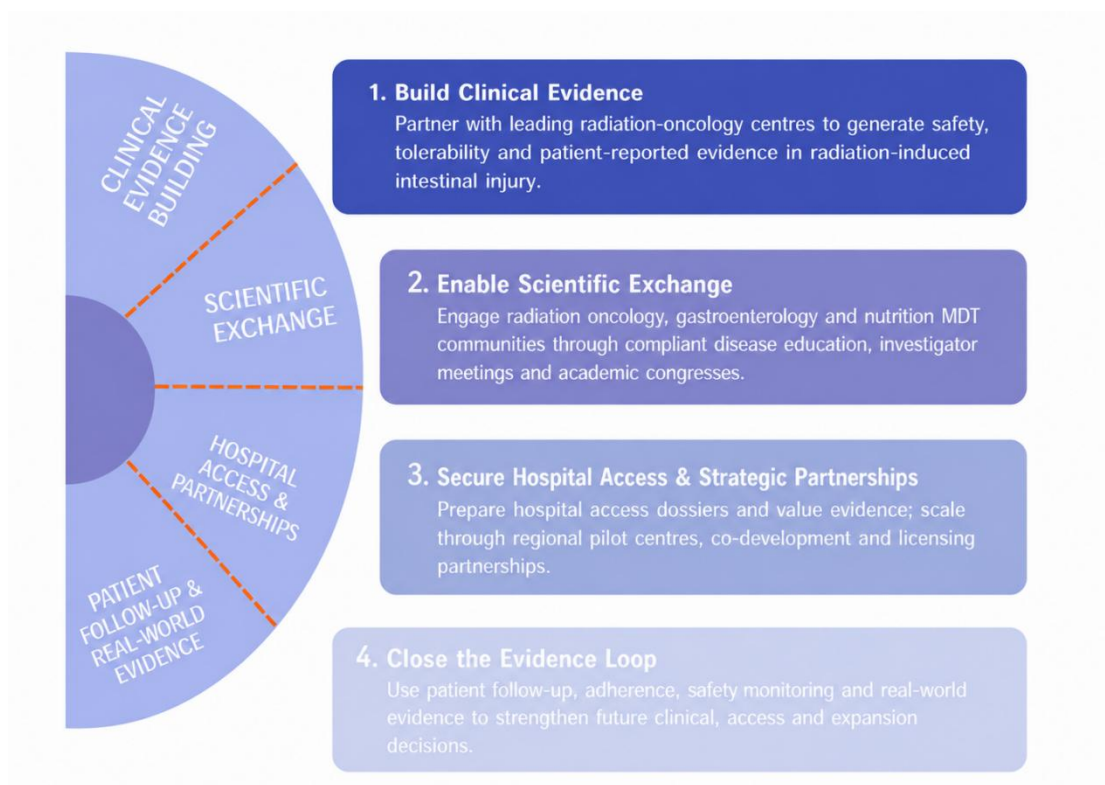


Figure 11 Medical-led Market Growth Strategy

8 MVP

8.1 MVP Design

MVP (Minimum Viable Product) refers to forming a testable product prototype with the most core functions in the early stage of product development and validating product assumptions at lower cost and in a shorter cycle. Its core value lies in prioritizing validation of the product's basic efficacy, delivery feasibility, and safety controllability before investing in large-scale preclinical research and industrialization development, thereby reducing subsequent R&D risks.

Based on the project's current R&D foundation, the MVP of Sybio-Gutweaver is designed as lyophilized engineered EcN + gut-release capsules. The candidate engineered bacteria integrate a DHICA-component-optimized melanin antioxidant module, a lactate barrier-support module, an Ag43 controllable adhesion module, and dual biosafety switches; Ag43 is used to enhance short-term retention in the target intestinal segment, and the Qwitch and L-arabinose-dependent hok/sok systems are used to limit persistent functional output and unintended dissemination under abnormal conditions.

According to suggestions from Ge Yin, Director of the Science and Technology Finance Department and related staff in science and technology finance and

achievement translation at Lanzhou Science and Technology Market, we will focus on validating three questions in the MVP stage: whether the engineered bacteria can alleviate radiation-related intestinal epithelial injury; whether lyophilized bacterial cells can recover activity and function after passing through a simulated gastrointestinal environment; and whether the engineered bacteria can simultaneously stop adhesion and actively exit after withdrawal of the exogenous license.

8.2 MVP Validation

This project evaluates the MVP from two dimensions: first, validating the core functions, oral delivery capability, and safety control mechanisms of the engineered bacteria through in vitro experiments; second, evaluating the clinical demand and further translation value of the product through feedback from hospitals, research institutions, and industrial partners.

8.3 Preliminary Cooperation Intentions

In addition to MVP technical validation, the project simultaneously evaluates actual application needs through clinical and industrial exchanges. At present, the team has conducted exchanges with Gansu Provincial Hospital, the Heavy Ion Center of Gansu Wuwei Tumor Hospital, the Institute of Modern Physics of the Chinese Academy of Sciences, and industry experts related to live bacterial formulations, preliminarily validating the clinical demand for radiotherapy-related intestinal injury, engineered live bacterial delivery schemes, and subsequent product translation directions.

9 Financial Planning

9.1 Cost Assumptions

To ensure that Sybio-Gutweaver advances from an engineered bacterial prototype to a candidate product, preclinical research, and early clinical stages, this project establishes a cost model based on the R&D characteristics of engineered live bacterial drugs. The project adopts an asset-light model of “independent R&D + third-party CRO/CDMO + co-development/technology licensing,” focusing on mastering core strains, functional circuits, safety control, delivery systems, and intellectual property, while prioritizing cooperation with pharmaceutical companies for late-stage clinical development, scaled manufacturing, and commercialization.

1. Human Resource Costs

The LZU-Northwest team currently has 28 members. Combined with post-incorporation position needs, approximately 12 core members are expected to

participate full-time in subsequent R&D and commercial translation, covering functions such as strain R&D, formulation and CMC, regulatory quality, clinical cooperation, intellectual property, business development, and operations.

Data from the National Bureau of Statistics show that in 2025, the average wage of employees in “scientific research and technical services” in urban non-private units nationwide was RMB 182,064 per year. Considering that the project is in an early entrepreneurial stage and adopts a compensation model of “cash compensation + long-term equity incentives”, this project uses RMB 180,000 per person per year as the average cash compensation of the core team. Including social insurance, housing provident fund, and other employee benefits, and estimating actual employer cost at 1.30 times cash wages, annual human resource expenditure is about RMB 2.81 million.

2. Training Expenditure

Engineered live bacterial R&D involves professional requirements such as biosafety, GMP/GCP, drug registration, quality systems, and clinical research. This project sets a training budget of RMB 8,000 per person per year, with cumulative training costs of about RMB 576,000 over six years. Including personnel compensation, training, recruitment, and necessary employee support costs, the six-year “team building” budget is accrued at RMB 18 million.

3. Third-Party Services

Sybio-Gutweaver is an engineered live bacterial candidate drug, and some regulatory registration, CMC, preclinical safety evaluation, and clinical research need to be entrusted to professional institutions.

(1) Legal, regulatory, and quality systems. The project involves matters such as engineered live bacterial R&D, biosafety, drug registration, clinical research, and license-out, requiring continuous support for registration strategy, submission materials, quality systems, third-party testing audits, and legal contracts. At present, the registration fee for a drug clinical trial application for a new drug produced domestically is RMB 192,000 per product. Including registration, quality, and legal compliance expenditures, the six-year regulatory and quality system budget is accrued at RMB 6 million, which includes related legal and professional consulting fees and is not calculated separately again.

(2) CMC and formulation development. Sybio-Gutweaver uses “lyophilized engineered EcN + gut-release capsules” and needs to complete strain banks, fermentation, lyophilization, viable bacterial potency, genetic stability, release performance, quality control, stability, and clinical sample manufacturing. Product validation also explicitly includes moisture content, lyophilization recovery rate, survival in simulated gastric fluid, release in simulated intestinal fluid, and functional

recovery after release.

In public government procurement, the budget for a preclinical CMC study supporting IND was RMB 3.1 million. Considering additional strain identification, viable bacterial potency, lyophilization recovery, and biosafety evaluation for engineered live bacteria, this project estimates the budget for complete CMC, formulation development, and clinical samples at RMB 12 million.

(3) Preclinical safety evaluation. In addition to conventional toxicology, engineered live bacteria also require evaluation of in vivo distribution, bacterial shedding and clearance, genetic stability, safety switch escape, and potential environmental release. In government procurement, the budget for a nonclinical pharmacokinetic, toxicokinetic, and safety evaluation project supporting IND submission was about RMB 3.4 million, with a winning bid of RMB 3.36 million. Considering the live bacterial-specific evaluation of Sybio-Gutweaver, the preclinical safety evaluation budget for this project is about RMB 6 million.

(4) Clinical trials. A first-in-human (FIH) Phase I clinical study publicly disclosed in 2026 had a budget of RMB 9 million. Sybio-Gutweaver also needs additional testing for bacterial shedding, strain recovery and clearance, and abnormal colonization; therefore, the Phase I/early human study budget for this project is estimated at RMB 12 million. Phase II/III will in principle be jointly undertaken by subsequent co-development or license-out partners and is not included in the base budget at this stage.

(5) Professional consulting and financial services. The project will engage external professional institutions as needed for registration strategy, industrial translation, financing, and fiscal and tax management, with related costs arranged annually according to R&D and financing stages. Among these, registration and industrial consulting expenditures are included in the regulatory and quality system budget, while fiscal, tax, and basic accounting services are included in daily operating expenditures and are not separately set as cost items.

4. Infrastructure and R&D Operations

In the early stage, the project will continue to use universities and partner research platforms and will not build an independent commercial production line. Related costs mainly include core strain R&D, animal efficacy, experimental platforms, small instruments, consumables, software, sample transportation, and R&D repetition and reserve costs.

The six-year “core R&D and productization” budget totals RMB 50 million, including about RMB 12 million for core experimental R&D, about RMB 12 million for CMC and clinical samples, about RMB 6 million for preclinical safety evaluation, about RMB 12 million for early clinical work, and the remaining about RMB 8

million for R&D operations, animal experiments, platform use, and R&D reserve expenditures.

5. Intellectual Property and Industrial Cooperation

The project will form portfolio intellectual property around the engineered EcN chassis, DHICA-optimized circuit, lactate module, Ag43 controllable adhesion, Qwitch + hok/sok dual safety mechanism, and lyophilization-intestinal release system, and will advance PCT, FTO, and subsequent patent layout according to commercial value, with a six-year intellectual property budget accrued at RMB 4 million.

Pre-approval market investment is mainly used for hospital cooperation, communication with clinical experts, medical conferences, industrial BD, and license-out preparation. At the current stage, the project has conducted preliminary demand and product direction validation through hospitals, heavy ion centers, research institutions, and live bacterial formulation industry experts. The six-year hospital and industrial cooperation budget is accrued at RMB 5 million.

Cost Summary

Based on the above assumptions, Sybio-Gutweaver's estimated baseline cash expenditure for 2026–2031 is RMB 83 million.

Table 14 2026-2031 Budget Summary

Cost Category	Six-Year Budget (RMB 10,000)
Core R&D and productization	5,000
Regulatory and quality system	600
Intellectual property	400
Team building	1,800
Hospital and industrial cooperation	500
Total	8,300

According to R&D stages, annual expenditures are estimated as follows:

Table 15 2026-2031 Annual Expenditure Budget

Year	Core R&D and Productization	Regulatory and Quality System	Intellectual Property	Team Building	Hospital/Industrial Cooperation	Total
2026	230	30	100	300	40	700
2027	620	60	90	300	80	1,150
2028	820	80	70	300	80	1,350
2029	1,490	180	60	300	120	2,150
2030	1,230	170	50	300	100	1,850
2031	610	80	30	300	80	1,100
Total	5,000	600	400	1,800	500	8,300

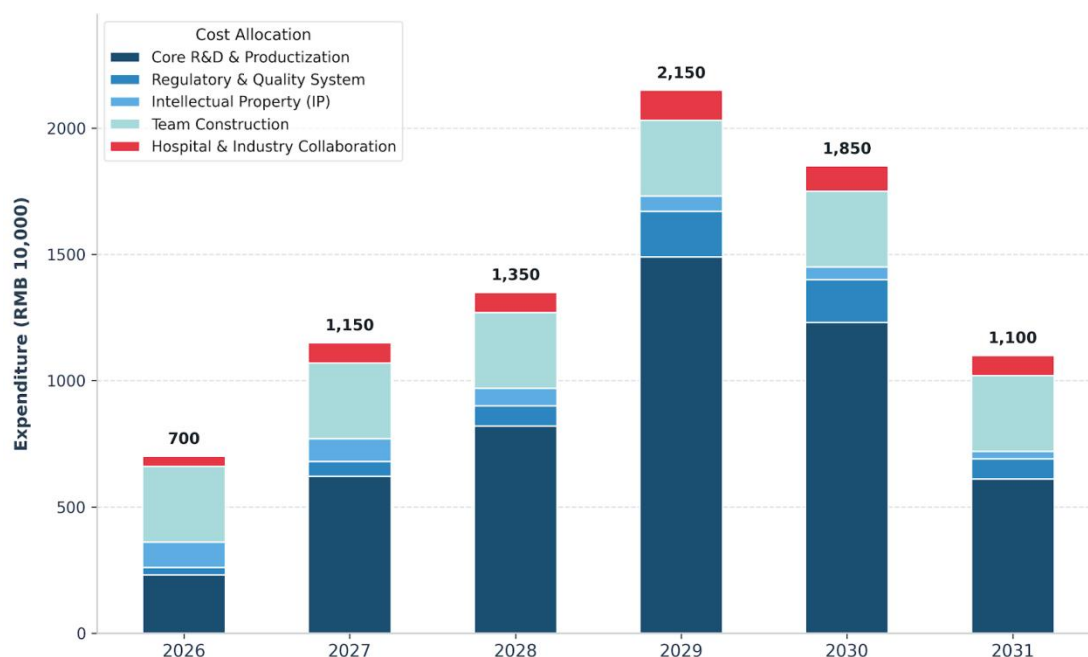


Figure 12 2026-2031 Cost Summary

9.2 Revenue Assumptions

Sybio-Gutweaver adopts an asset-light business model of “independent R&D + co-development/technology licensing.” During the forecast period, it will not rely on independent drug sales as the main revenue source, but will mainly realize commercial returns through co-development payments, license-out upfront payments, R&D milestone payments, and post-launch sales revenue sharing.

There are already corresponding licensing cases in the live bacterial and microecological therapy field. Synlogic’s engineered bacteria collaboration with AbbVie included USD 2 million upfront and up to USD 16.5 million in R&D milestones; Seres’ collaboration with Nestlé Health Science around SER-109 included USD 175 million upfront, USD 125 million in FDA approval milestones, and up to USD 225 million in sales milestones. These cases indicate that the clinical maturity and regulatory certainty of an asset significantly affect licensing value.

Considering that Sybio-Gutweaver is still at an early stage, this project adopts clearly more conservative internal revenue assumptions:

Table 16 Internal Revenue Assumptions

Revenue Source	Trigger Condition	Baseline Assumption
Co-development/Option payment	reproducible efficacy + preliminary CMC	RMB 5 million
License-out upfront payment	Reaching IND-ready or key submission milestone	RMB 20 million
Development Milestone I	Phase I completed	RMB 10 million
Development Milestone II	Phase II initiated	RMB 20 million

Sales revenue sharing	Product commercialization	5% of partner net sales
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The above amounts are only used for business plan calculations and do not represent signed transaction quotations. If the Option and formal License-out come from the same partner, the earlier Option payment may offset part of the upfront payment according to the final agreement.

Considering uncertainty in license-out signing timing and payment conditions, the 2026–2031 baseline cash flow does not recognize the above licensing revenue in advance, and recognizes it only after a formal cooperation agreement is signed and corresponding payment conditions are met.

9.3 Financing Plan

Innovative drug projects have long R&D cycles and lack stable revenue and fixed assets usable for large loans in the early stage; therefore, this project mainly relies on equity financing, supplemented by scientific research project funds and industrial cooperation income. Professional life science investment institutions can provide not only funds but also CMC, clinical, regulatory, and subsequent industrial cooperation resources.

9.3.1 Financing Benchmarks

In recent years, early-stage financing amounts for domestic innovative drugs have varied widely. In 2025, Hailai Ke completed nearly RMB 60 million in angel-round financing; Xinglian Peptide completed RMB 100 million in Pre-A+ financing; Yangguang Anjin completed more than RMB 100 million in Pre-A financing, with related funds mainly used for preclinical or early clinical development of candidate drugs.

Different technology routes and R&D stages cannot be directly compared horizontally, but public cases indicate that innovative drug financing scale mainly depends on candidate product maturity, clinical progress, and use of funds. By contrast, this project adopts a staged, milestone-driven financing plan and remains relatively prudent overall.

9.3.2 Financing Proposal

The project financing roadmap corresponds to R&D milestones and plans to complete three rounds of financing:

Angel round: RMB 10 million. Used for integration of core functional modules, validation of Ag43 and dual safety switches, lyophilized-gut-release capsule MVP, and core intellectual property layout.

Pre-A round: RMB 25 million. Used for Lead Strain convergence, reproducible animal efficacy, genetic and formulation stability, safety switch escape rate, critical

quality attributes, CMC process development, and registration preparation.

Series A: RMB 55 million. Used for registration-grade CMC, nonclinical safety evaluation, GMP clinical samples, clinical submission, and early clinical research, with a focus on advancing co-development and license-out.

The three rounds total RMB 90 million. Of this, RMB 83 million is used for baseline R&D and operating expenditures, and the remaining about RMB 7 million serves as a cash safety reserve to address extended R&D cycles, supplementary studies, and financing delays.

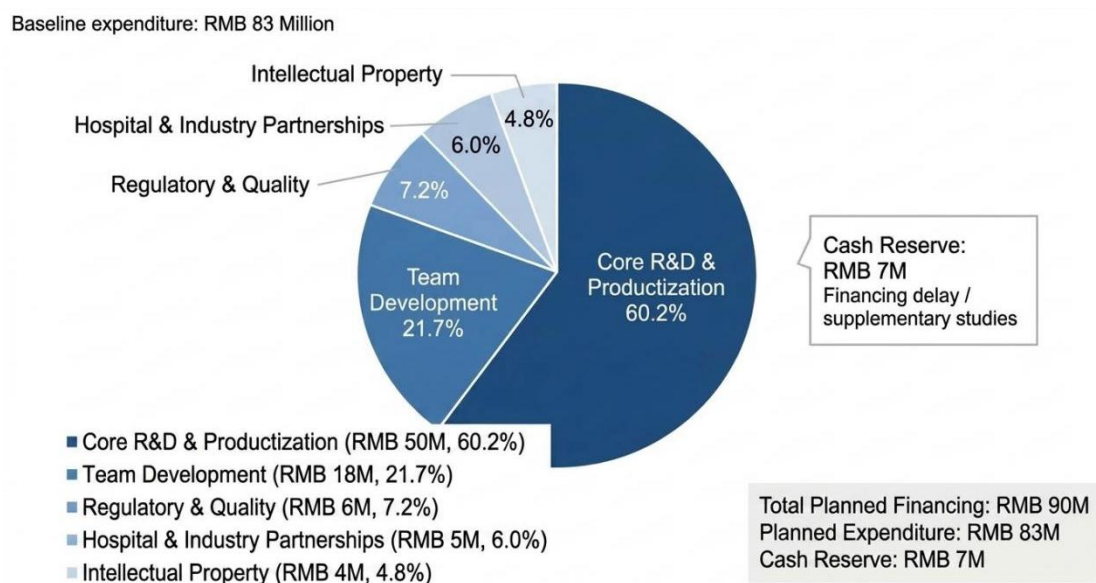


Figure 13 Proportion of Fund Usage

9.3.3 Corporate Valuation

Through in-depth communication with Wang Huanan, founder and chairman of Shenzhen Huanuo Biology, we believe that Sybio-Gutweaver has not yet formed stable product revenue, and there is considerable uncertainty in R&D success rate, launch timing, and transaction value; therefore, traditional DCF is not adopted as the main valuation method at this stage. Examples also indicate that the valuation of early innovative drug companies depends more on market environment, technological potential, and negotiations between investors and the company.

This project uses financing amount and target equity dilution ratio as internal valuation references:

Financing Round	Financing Amount	Target Dilution Ratio	Reference Pre-money
Angel round	RMB10million	12%–15%	RMB56.67 – 73.33 million
Pre-A	RMB25million	15%–18%	RMB 114 – 142 million
Series A	RMB55million	15%–20%	RMB 220 – 312 million

The above valuations are only references for financing plans and negotiations. Project valuation will mainly increase with key milestones such as Lead Strain,

reproducible animal efficacy, genetic and formulation stability, safety switch escape rate, CMC manufacturability, preclinical safety, and IND-ready status.

9.3.4 Target Financing Capital

The project prioritizes four types of capital: professional life science VCs, early-stage funds focused on synthetic biology and engineered bacterial technology, local biopharmaceutical industry funds, and CVCs of pharmaceutical companies with microecology or biologics R&D capabilities.

In screening investment institutions, not only fund size but also their follow-on financing, CMC/CDMO, clinical registration, and license-out resources should be considered. Industrial partners may be approached early in the Pre-A stage, but in principle formal licensing negotiations on core rights will begin only after relatively complete preclinical evidence is formed, so as to maintain the project's bargaining power.

9.4 Cash Flow Forecast

Based on the above cost and financing plans, this project conducts annual cash flow forecasts for 2026-2031. The baseline scenario includes only planned equity financing and does not recognize uncertain government subsidies, license-out, or milestone revenue in advance.

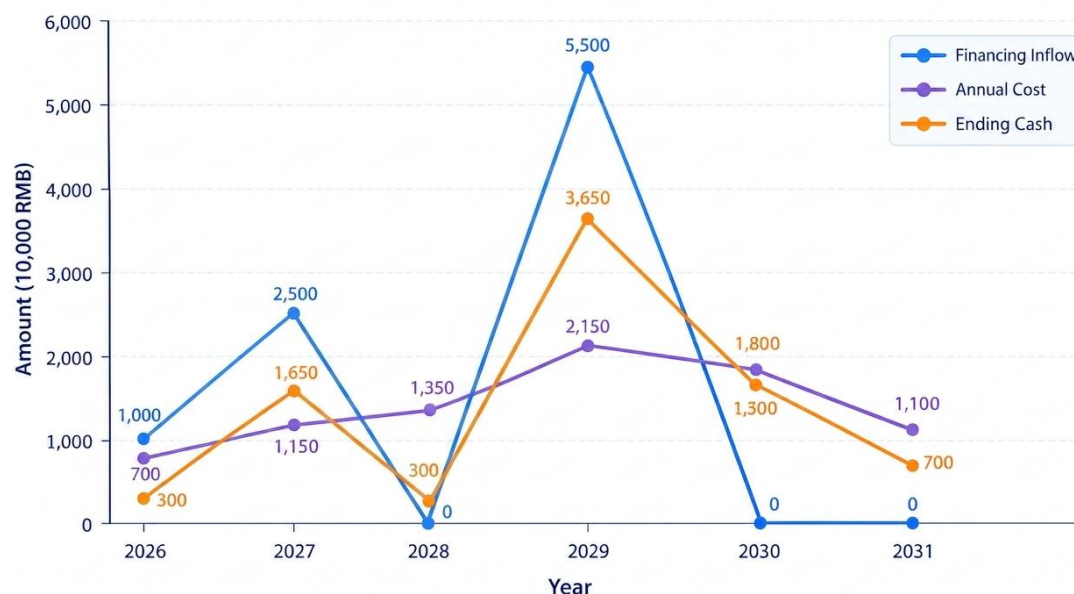


Figure 14 Cash Flow under Normal Financing Scenario

Under the normal financing scenario, if the angel round, Pre-A round, and Series A complete financing of RMB 10 million, RMB 25 million, and RMB 55 million respectively, cumulative financing over six years will be RMB 90 million, which can cover estimated R&D and operating expenditures of RMB 83 million and retain about

RMB 7 million in cash reserves by the end of 2031. Among these, cash at the end of 2028 is expected to decline to about RMB 3 million, the most critical financing window during the forecast period; therefore, Series A should be launched in advance while funds can still cover subsequent core R&D.



Figure 15 Cash Flow under 80% Financing Stress Scenario

If all three financing rounds reach only 80% of the planned amount, the project will have a funding gap of about RMB 4 million in 2028; after Series A funds arrive, the cash position may temporarily recover, but by 2031 a funding gap of about RMB 11 million is expected again, indicating that insufficient financing progress will significantly increase late-stage funding chain risk.

Therefore, when financing is below expectations, the project should launch subsequent financing in advance, appropriately defer non-core expenditures, and reduce short-term cash pressure through staged payments to CRO/CDMO. After stable candidate strains, preclinical evidence, and preliminary CMC data are formed, co-development or license-out can be advanced in parallel to supplement funds for subsequent clinical development.

10 Risk Control

10.1 Risk Interview

We conducted in-depth exchanges with Associate Professor Liu Zhaoyang from the School of Management of Lanzhou University and Associate Professor Li Hao

from the School of Economics of Lanzhou University, forming a more systematic understanding of project risk management, focusing on risk quantification, responses to funding shortages, and contingency plan formulation. Professor Liu Zhaoyang has long engaged in research on financial markets, investment management, and corporate capital operations, and has experience in investment project consulting. He pointed out that innovative enterprises face high uncertainty in technological R&D and achievement translation, and therefore need to identify key risks as early as possible and integrate risk management into corporate development and financing decisions.

First, Professor Liu Zhaoyang emphasized the importance of risk quantification. Different risks differ markedly in their impact on enterprise development and probability of occurrence, and qualitative judgment alone makes it difficult to determine management priorities. Therefore, risks such as R&D delays, approval progress, market changes, and cash flow can be quantified through risk assessment models and scenario analysis. Based on this suggestion, Sybio-Gutweaver conducts risk assessment from three dimensions-impact, probability, and uncontrollability-to identify key risks requiring priority monitoring and intervention.

Regarding funding shortages, Professor Liu Zhaoyang pointed out that innovative drug projects have long R&D cycles and large early investment, and cash flow management is particularly important before stable revenue is formed. Financing arrangements should match R&D milestones, and necessary cash reserves should be retained to address R&D delays and changes in financing progress. Based on this approach, Sybio-Gutweaver sets up staged financing through angel, Pre-A, and Series A rounds and reserves a cash safety buffer; if financing is delayed, priority will be given to core work such as candidate strains, CMC, biosafety validation, and registration preparation.

Professor Liu Zhaoyang further emphasized the importance of contingency plans. When facing uncertainties in R&D, financing, and the market, enterprises should formulate resource adjustment plans under different scenarios in advance. When the R&D cycle is extended or financing falls short of expectations, budgets and development pace should be adjusted in a timely manner, concentrating limited funds on core links that determine project value. At the same time, communication with investment institutions and industrial partners should be maintained, and financing and cooperation strategies should be dynamically adjusted according to project progress, thereby reducing the impact of a single risk on the overall development process.

Overall, this interview made us recognize that risk management is not only control of potential losses but also an important management tool for ensuring continuous project advancement. Sybio-Gutweaver will improve its adaptability to R&D and funding uncertainty through risk quantification, milestone financing, cash

reserves, and dynamic contingency mechanisms, providing support for subsequent clinical translation and industrial cooperation.

10.2 Risk Identification and Assessment

From candidate strain development to registration and industrialization, Sybio-Gutweaver mainly faces development window, biosafety, CMC and approval, market acceptance, cash flow, and ethical compliance risks. The project evaluates risks from three dimensions: Impact, Probability, and Uncontrollability, each using a 1–5 Likert scale. Risk Index = Impact × Probability × Uncontrollability. The higher the risk index, the more the risk requires priority monitoring and intervention.

This project classifies a risk index ≥64 as high risk, 27-63 as medium risk, and <27 as low risk. The higher the risk index, the more the risk requires priority monitoring and management.

CATEGORY	RISK	PROBABILITY	IMPACT	RISK RATING	SOLUTION
PRODUCT RISK	Lead Strain Instability	4	5	High	Set stage gates; verify genetic and functional stability.
	Insufficient Therapeutic Efficacy	3	5	High	Strengthen efficacy testing; optimize key functional modules.
	Biosafety Switch Escape	3	5	High	Validate dual safety switches; monitor escape and clearance rates.
	CMC and Batch Inconsistency	4	5	High	Strengthen CMC control; optimize freeze-drying and batch consistency.
	Slow Clinical Acceptance	3	4	Medium	Engage clinicians early; build clinical evidence and hospital access.
FINANCIAL RISK	Cash Flow Shortage	4	5	High	Plan milestone financing; control costs; maintain emergency reserves.
ETHICAL RISK	Biosafety and Ethical Controversy	2	5	High	Strengthen ethical review; monitor safety; establish reporting mechanisms.
STRATEGIC RISK	Missing Development Window	4	5	High	Focus on Lead Strain; monitor market and accelerate development.
REGULATORY POLICY RISK	Regulatory Pathway Changes	3	5	Medium	Monitor policies; communicate with regulators; conduct compliance reviews.
MANAGEMENT RISK	Cross-Functional Coordination Failure	2	4	Low	Clarify responsibilities; strengthen cross-review and project management.
SECURITY RISK	IP and Data Leakage	2	4	Medium	Strengthen access control, NDAs, data security and version management.

Figure 16 Risk Identification and Assessment

For matters with potentially severe consequences, such as safety switch failure, abnormal long-term persistence of engineered bacteria, environmental escape, and major subject safety events, the project establishes a mandatory escalation mechanism for major risks. Such matters are independent of the risk index; once triggered, they are managed as the highest-level risk, related activities are suspended, and continuation is permitted only after investigation, correction, and independent review.

10.3 Risk Management Measures

Tiered management system. High-risk matters undergo monthly management review and immediate intervention upon reaching preset trigger conditions; medium- and low-risk matters are updated quarterly. Risk management is also linked to Go/No-Go decisions at key nodes such as Lead Strain, CMC, and registration preparation.

Segregation of duties strategy. Strain R&D, CMC and quality, regulatory and biosafety, and business and finance are respectively responsible for R&D, quality, compliance, and financing translation; key safety and quality conclusions undergo cross-review to reduce judgment bias from a single department.

Collaborative risk-sharing strategy. The project conducts demand validation, quality systems, and subsequent translation jointly with hospitals, research institutions, and professional service agencies; in intellectual property, it establishes a management chain of “NDA-MTA-joint R&D-achievement ownership-License”, and after evidence matures, shares late-stage risks with pharmaceutical companies through co-development or license-out. In industrial consulting, Ge Yin and Qi Xiaoyu of Lanzhou Science and Technology Market pointed out that investors and industrial partners focus more on stability, safety, batch consistency, and intellectual property boundaries than on single best experimental data or the number of modules.

Policy and strategic utilization. The project adopts “NMPA primary pathway, FDA parallel assessment”, conducts regulatory communication at key nodes, and reserves an FDA LBP registration interface.

Commercialization adheres to “ground-based radiotherapy first, parallel international market layout, and long-term extension to space scenarios,” prioritizing the formation of a clinical and commercial closed loop in radiotherapy-related intestinal injury scenarios with a clearer patient base and validation pathway, and then extending to space scenarios.

10.4 Special Response Measures for High-Risk Matters

10.4.1 Development Window Risk

Candidate strain stage gate. R&D focus shifts from continuously stacking

functional modules to Lead Strain convergence, with emphasis on validating functional reproducibility, genetic stability, lyophilization stability, safety switch performance, and manufacturability.

Module de-escalation mechanism. If a module significantly increases metabolic burden, manufacturing complexity, or safety risk without generating corresponding clinical value, its expression intensity will be reduced, its control mode adjusted, or integration temporarily deferred.

Rapid market feedback mechanism. Continuously validate demand and acceptance through radiotherapy, oncology, pharmacy experts, and key hospitals, and track competitive progress in engineered live bacteria and radiotherapy supportive care, avoiding disconnection between R&D direction and market demand.

10.4.2 Approval Risk

Front-loading of registration pathway. The project advances under the positioning of a therapeutic engineered live bacterial drug, with NMPA as the main pathway, conducts CDE communications at key nodes, and reserves an international submission interface under the FDA LBP framework.

CMC compliance pre-assessment. Establish quality control systems in advance for strain identification, fermentation, lyophilization, viable bacterial potency, genetic stability, product stability, and batch consistency, so that product design and registration requirements advance in parallel.

Policy early-warning mechanism. Continuously track engineered live bacteria, biosafety, and drug registration policies, and conduct simultaneous regulatory and intellectual property assessments before major strain, dosage form, or process adjustments.

10.4.3 Product-Specific Risk

Ag43 and survival license linked control. Ag43 is used to achieve controllable short-term retention in the target intestinal segment rather than long-term colonization. The project binds Ag43 expression and hok/sok survival licensing to the same exogenous L-arabinose license signal: when L-arabinose is present, sok maintains bacterial survival and allows Ag43 expression; after the license signal disappears, the toxic effect of hok dominates while Ag43 expression stops, thereby reducing the risk of persistent adhesion after loss of control.

Dual safety control. In early R&D iterations, the project has formed a two-layer safety logic around “exit under abnormal microenvironment + survival under exogenous license”: Qwitch triggers self-destruction in a high-lactate environment, and the L-arabinose-dependent hok/sok system controls engineered bacterial survival and Ag43 expression, forming a complementary pair.

Pre-registration safety indicators. Before entering subsequent registration and clinical translation preparation, quantitative validation should be completed for safety switch trigger thresholds, response time, killing efficiency, escape rate, genetic stability, and Ag43 shutdown effect. If preset standards are not met, the project will not enter the next development stage. Legal experts also pointed out that merely proving the “existence of a safety switch” is insufficient to form adequate safety evidence.

10.4.4 Cash Flow Risk

Milestone financing mechanism. The angel round, Pre-A round, and Series A correspond respectively to MVP, Lead Strain convergence, and registration and early clinical stages, matching financing pace with project value appreciation.

Early launch of financing. Under the normal scenario, cash at the end of 2028 is about RMB 3 million; if financing reaches only 80% of the planned amount, a funding gap of about RMB 4 million may occur in 2028, so subsequent financing should be launched in advance.

Flexible budget and cash reserves. Planned financing is RMB 90 million, of which RMB 83 million is used for baseline R&D and operations and RMB 7 million serves as a cash safety reserve. If financing is delayed, priority will be given to Lead Strain, CMC, biosafety validation, and registration preparation, while non-core and space expansion expenditures will be deferred.

10.4.5 Ethical Risk

Subject protection mechanism. When future human research is involved, the properties of engineered live bacteria, abnormal persistence, bacterial shedding, and potential unknown risks should be fully explained, and informed consent, privacy protection, and voluntary withdrawal systems should be strictly implemented.

Adverse event and independent monitoring mechanism. Establish AE/SAE recording, reporting, tracking, and emergency response systems. Set up an independent data safety monitoring board/independent data monitoring committee (DSMB/IDMC) according to research risk and clinical protocol to independently evaluate cumulative safety data; if abnormal persistent bacterial shedding, safety switch abnormalities, or other major safety signals occur, related research should be suspended in a timely manner and a special review initiated.

Independent ethics and conflict-of-interest management. Establish a three-line management system of “biosafety-science and technology ethics—human research ethics,” implement staged review for in vitro research, sample research, and future human research, and establish systems for original records, version management, and conflict-of-interest declaration. In project consulting, Mao Jian, a

biosafety and ethics lawyer at Lanzhou Leinuo Law Firm, pointed out that biosafety and ethics management of engineered live bacteria should not intervene only at the clinical stage but should run through the entire process of R&D, registration, and subsequent translation.

11 CCiC and Industry Collaboration

11.1 CCiC Background and Project Participation Positioning

CCiC (Conference of China iGEMer Community) is an important annual exchange platform for Chinese iGEM teams. For Sybio-Gutweaver, its core value lies in testing application assumptions easily overlooked in engineering design through external exchange. On July 25-26, 2026, the LZU-Northwest team went to Dalian University of Technology to participate in SynBioPunk-CCiC 2026 and won the Best Roadshow Award.

This project uses engineered EcN as the chassis, exploring the use of melanin to alleviate radiation-related oxidative stress and lactate to support the intestinal epithelial barrier, using oral administration as lightweight auxiliary protection for long-term deep-space missions, while evaluating its application potential in ground-based radiation intestinal injury. Because the project simultaneously involves functional output, intestinal action site, and live bacterial safety control, the team regards CCiC as a key external validation node in moving from “functionally achievable” to “systemically usable.”



11.2 Value of Project Participation in CCiC

The core value obtained by this project through CCiC exchange is mainly reflected in the following three aspects:

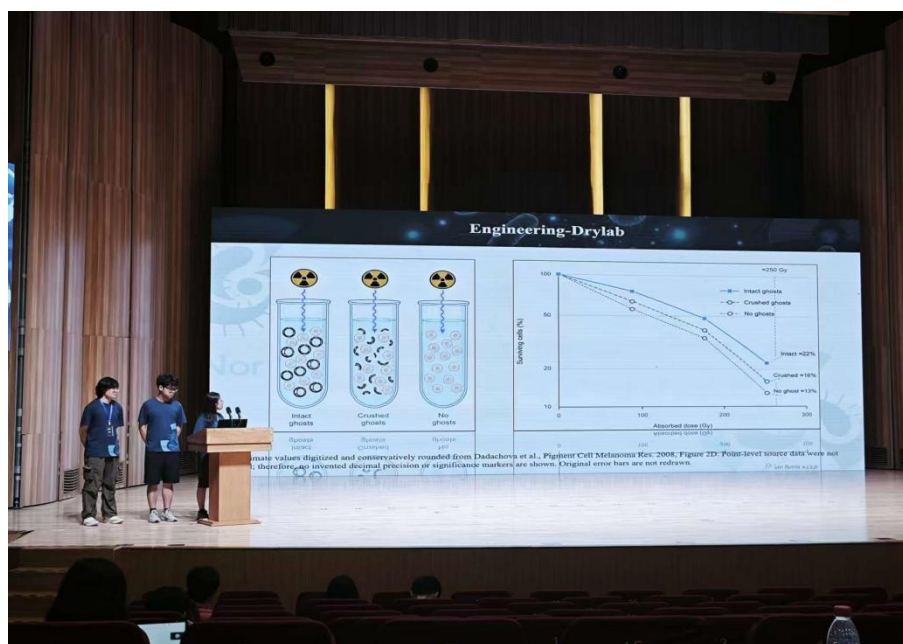
Validation of key assumptions: First, other teams and experts pointed out that even if engineered bacteria can produce target functional molecules, if they cannot

form sufficient local retention in the target intestinal region, the actual effect may still be limited. This feedback exposed the previous implicit premise that “reaching the intestine is sufficient to exert effects.”

Engineering parameterization: On this basis, the team added the Ag43 adhesion module and incorporated HIEC adhesion rate, residual after elution, bacterial aggregation, cytotoxicity, and expression stability into validation, transforming “intestinal retention” from a default condition into a measurable and controllable engineering parameter.

Front-loading of safety boundaries: Further analysis found that higher adhesion does not necessarily represent a better product. Ag43 may prolong local action time but may also increase unintended persistence; therefore, the project goal was adjusted from “enhanced colonization” to “controllable short-term retention,” and it is required to be linked with a safety exit mechanism.

During the project exchange session, Gao Rencheng, co-founder and CEO of JiYin Zaowu, affirmed the project’s innovative ideas and application value, and suggested that the team further improve the technical scheme, expand application scenarios, and enhance subsequent translation potential.



11.3 CCiC's Role in Advancing the Commercialization Path

If the problems exposed at CCiC cannot be solved early, they will translate into productization risks in preclinical evaluation, CMC development, and partner due diligence, mainly in the following three aspects:

Front-loading of product definition. Local retention determines the actual exposure window of functional molecules. If retention degree, duration, and clearance method cannot be demonstrated, subsequent efficacy differences will be difficult to

distinguish as insufficient function or insufficient delivery, and it will also be difficult to form stable candidate product specifications.

Upgrading of commercialization evidence. The project cannot merely prove that Ag43 “increases adhesion”; it also needs to form combined evidence on adhesion, elution, cytotoxicity, expression stability, and safety exit, and link it with lyophilization recovery, acid resistance, and intestinal release results. Ge Yin and Qi Xiaoyu of Lanzhou Science and Technology Market pointed out that investors and pharmaceutical companies focus more on reproducibility, stability, safety, and manufacturability than on a single best experimental result.

Clear cooperation interfaces. CCiC enabled the team to further clarify that subsequent external cooperation should focus on specific tasks such as intestinal model validation, live bacterial formulation development, and safety evaluation, rather than remaining at general project exchanges, thereby preparing data interfaces in advance for preclinical research, CMC processes, and subsequent technical due diligence.

11.4 Future Industry Cooperation and Development Prospects

Based on the problem list formed at CCiC, the project plans to further advance the following cooperation:

Conduct small-scale trials with relevant enterprises on lyophilization recovery, gastric acid protection, intestinal release, and batch consistency, so that strain function and product delivery are validated simultaneously.

Conduct radiotherapy-related intestinal injury models and safety clearance evaluation with hospitals and research institutions with existing clinical cooperation foundations, such as Gansu Provincial Hospital and the Heavy Ion Center of Gansu Wuwei Tumor Hospital, to establish evidence closer to real-world use scenarios for subsequent preclinical research.

CCiC is not only an important exchange platform for this project to receive external validation but also an important node in promoting the translation of engineering design from laboratory prototype to developable product. The project will rely on the problem feedback and exchange channels formed at CCiC to focus on advancing local retention model validation, live bacterial formulation processes, and preclinical safety evaluation, accelerating the translation of Sybio-Gutweaver from engineering validation to preclinical development and industrial cooperation.